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(54) **EXECUTION OF A MEMORY ACCESS OPERATION INVOLVING MULTIPLE MEMORY CELLS OF A MEMORY DEVICE USING MULTIPLE SENSE MODULES**

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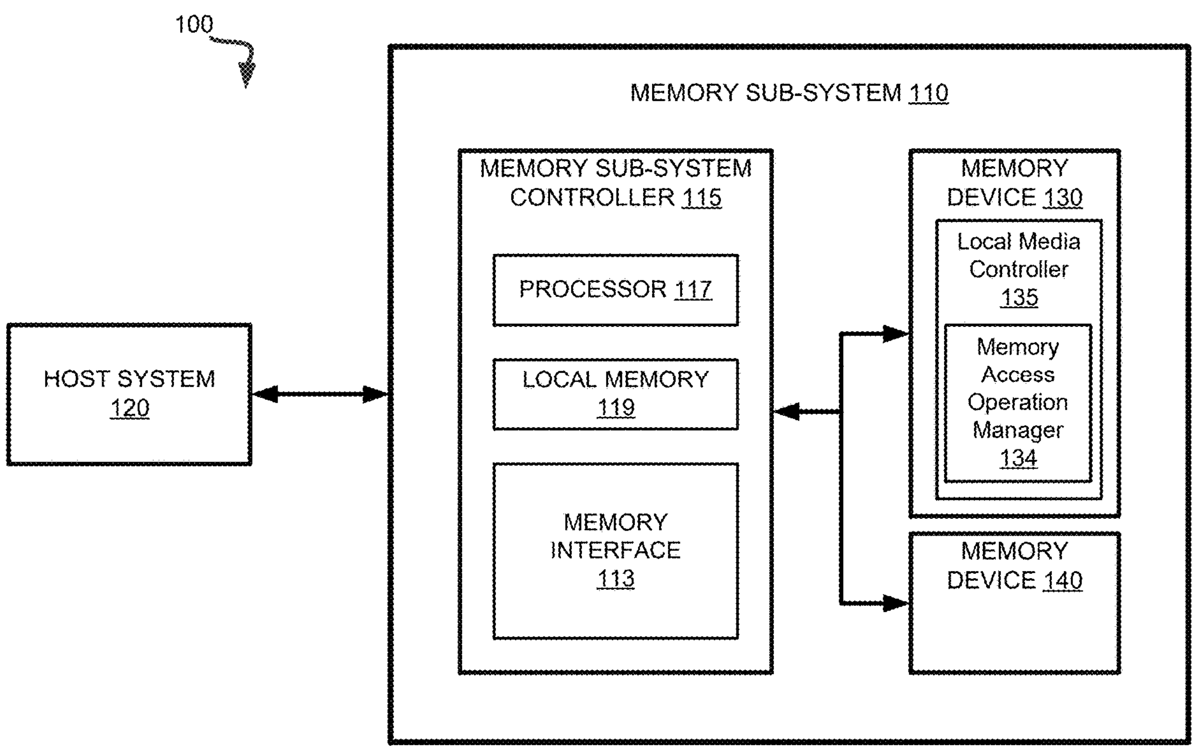
(57) **ABSTRACT**

Control logic in a memory device initiates a memory access operation associated with a memory block of a memory device, wherein the memory block includes a first subset of sub-blocks and a second subset of sub-blocks. During the memory access operation, the control logic causes transmission, via a first sense module coupled to the first subset of sub-blocks, of first data between a page buffer circuit and a first memory cell of the first subset of sub-blocks. During the memory access operation, the control logic causes transmission, via a second sense module coupled to the second subset of sub-blocks, of second data between the page buffer circuit and a second memory cell of the second subset of sub-blocks.



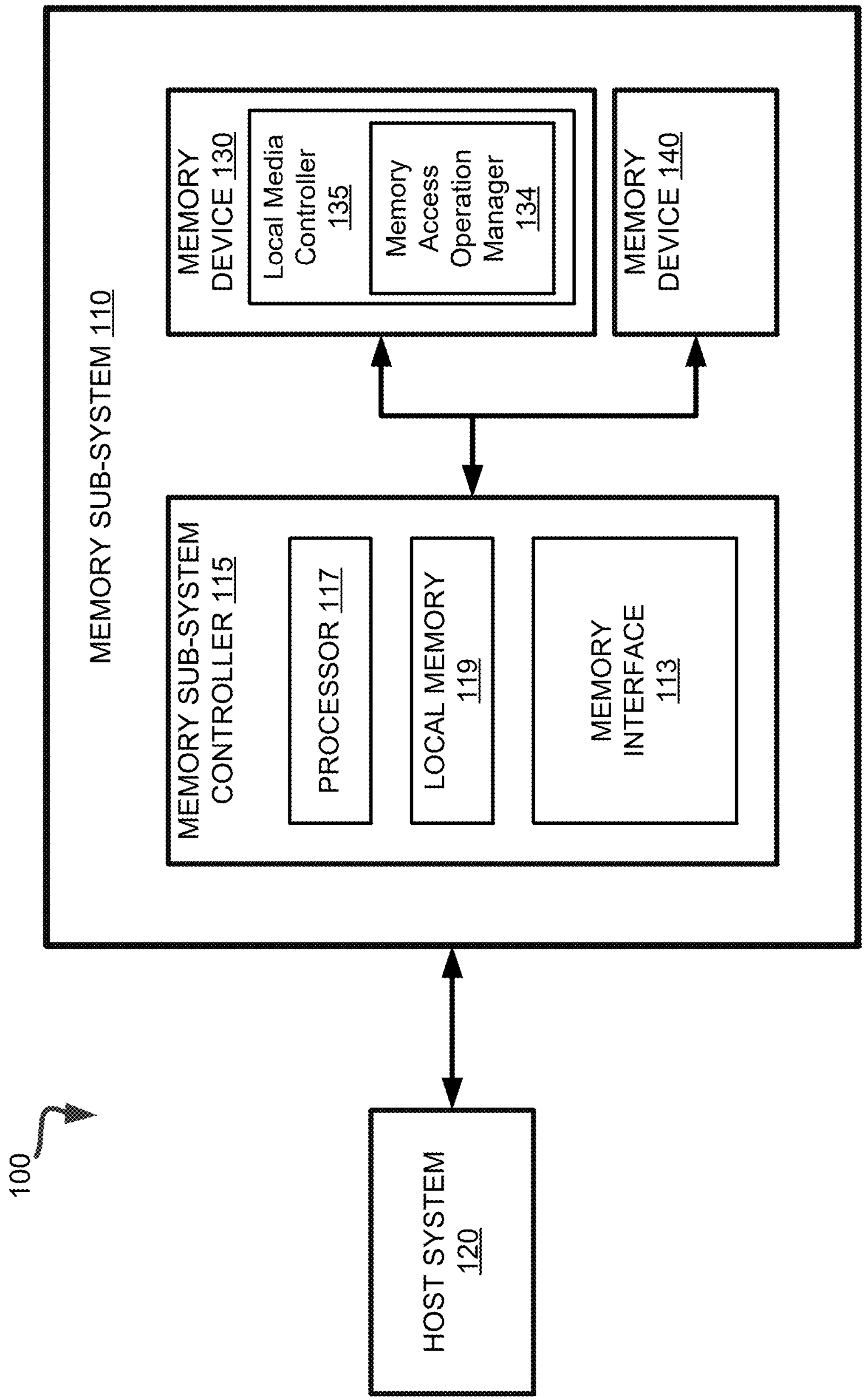

100
MEMORY SUB-SYSTEM 110
MEMORY DEVICE 130
Local Media Controller 135
Memory Access Operation Manager 134
MEMORY DEVICE 140
MEMORY SUB-SYSTEM CONTROLLER 115
PROCESSOR 117
LOCAL MEMORY 119
MEMORY INTERFACE 113
HOST SYSTEM 120

FIG. 1A

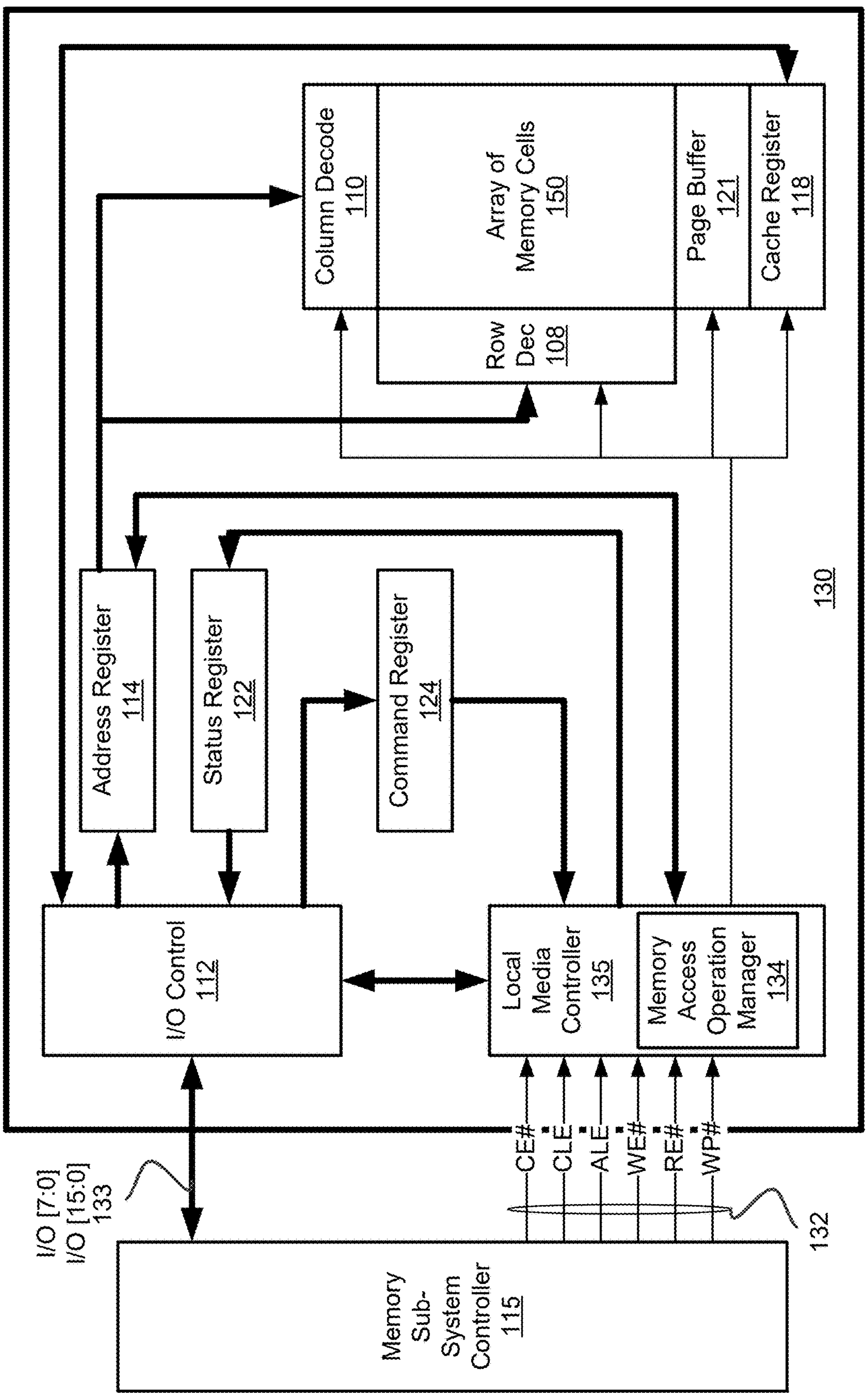

Column Decode 110
Array of Memory Cells 150
Page Buffer 121
Cache Register 118
Row Dec 108
Address Register 114
Status Register 122
Command Register 124
130
I/O Control 112
Local Media Controller 135
Memory Access Operation Manager 134
I/O [7:0]
I/O [15:0]
133
CE#
CLE
ALE
WE#
RE#
WP#
132
Memory Sub-System Controller 115

FIG. 1B

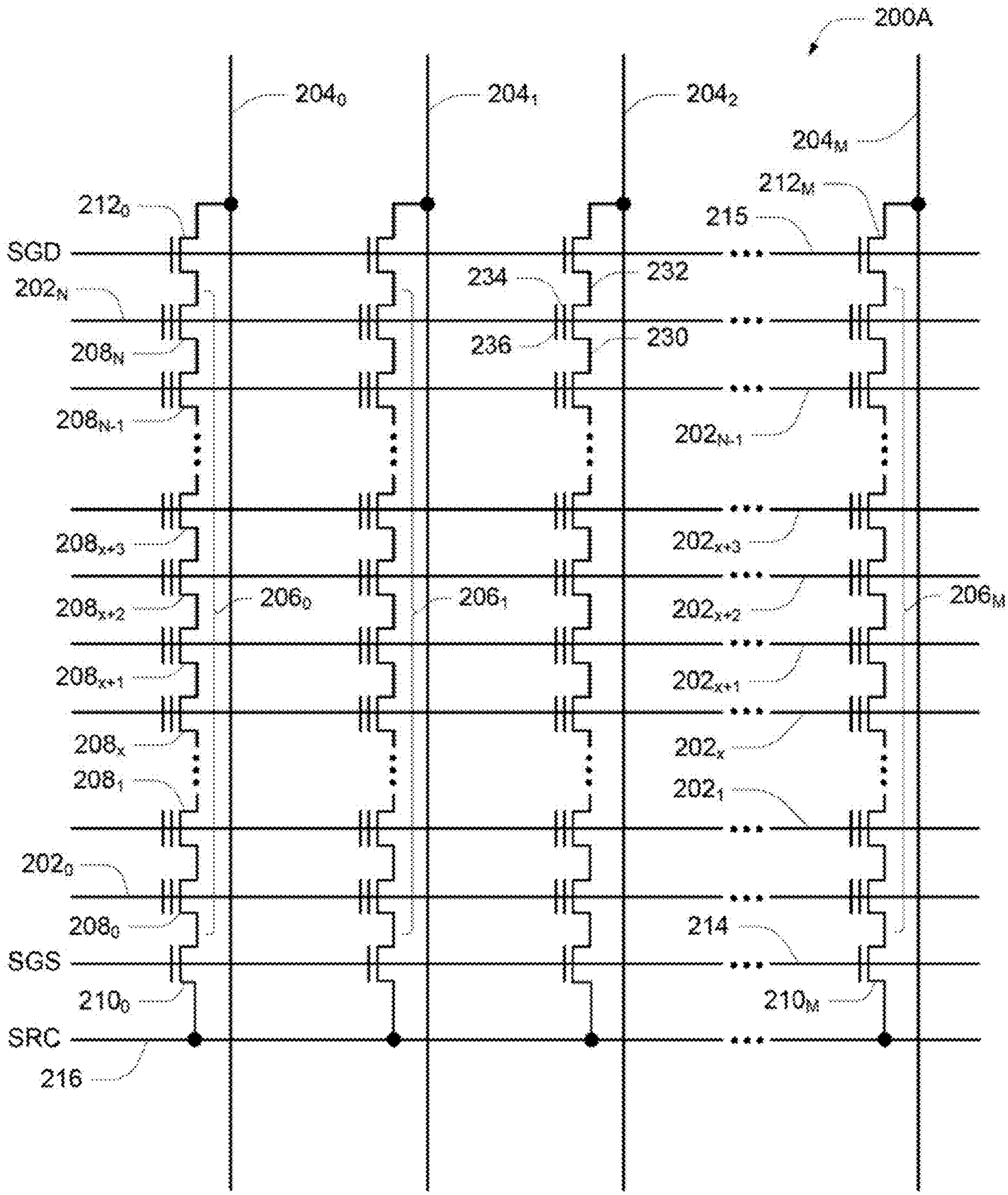

200A
204_0
204_1
204_2
204_M
212_0
212_M
215
SGD
202_N
234
232
236
230
208_N
208_{N-1}
202_{N-1}
208_{x+3}
202_{x+3}
208_{x+2}
206_0
206_1
202_{x+2}
206_M
208_{x+1}
202_{x+1}
208_x
202_x
208_1
202_1
202_0
208_0
214
SGS
210_0
210_M
SRC
216

FIG. 2A

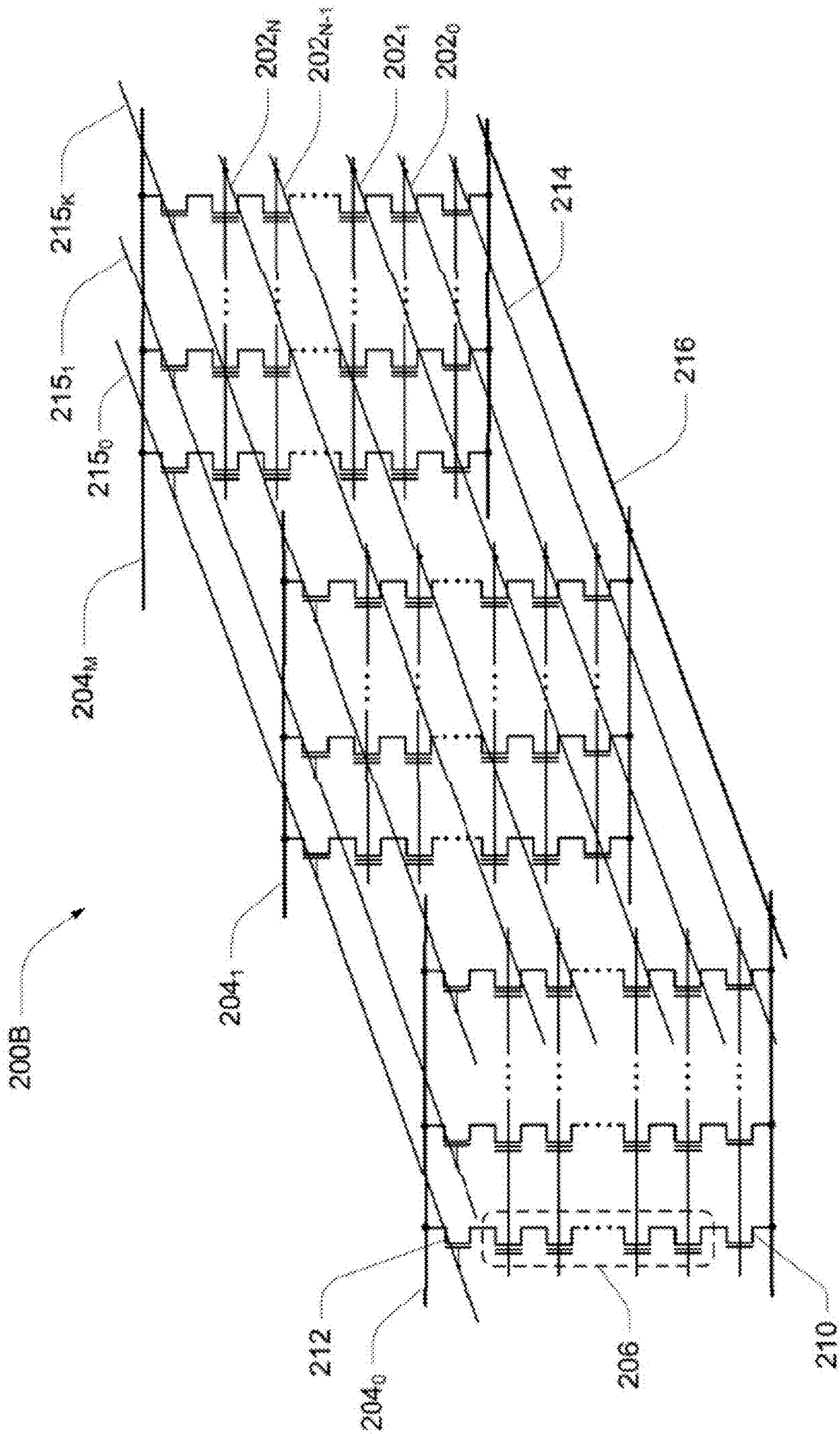
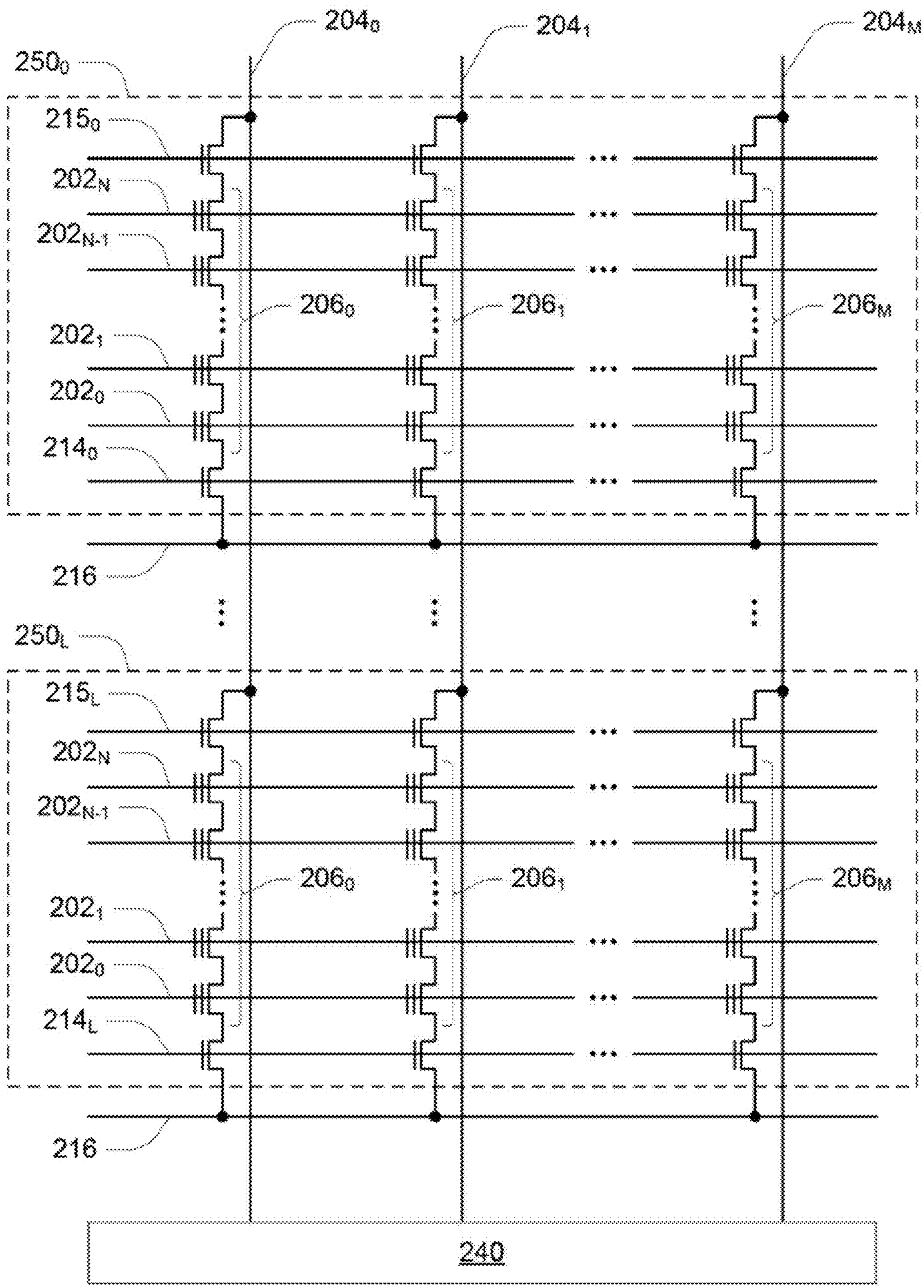

200B
204_0
212
206
210
204_1
204_M
215_0
215_1
215_K
202_N
202_{N-1}
202_1
202_0
214
216

FIG. 2B



204_0
204_1
204_M
250_0
215_0
202_N
202_{N-1}
206_0
206_1
206_M
202_1
202_0
214_0
216
250_L
215_L
202_N
202_{N-1}
206_0
206_1
206_M
202_1
202_0
214_L
216
240

FIG. 2C

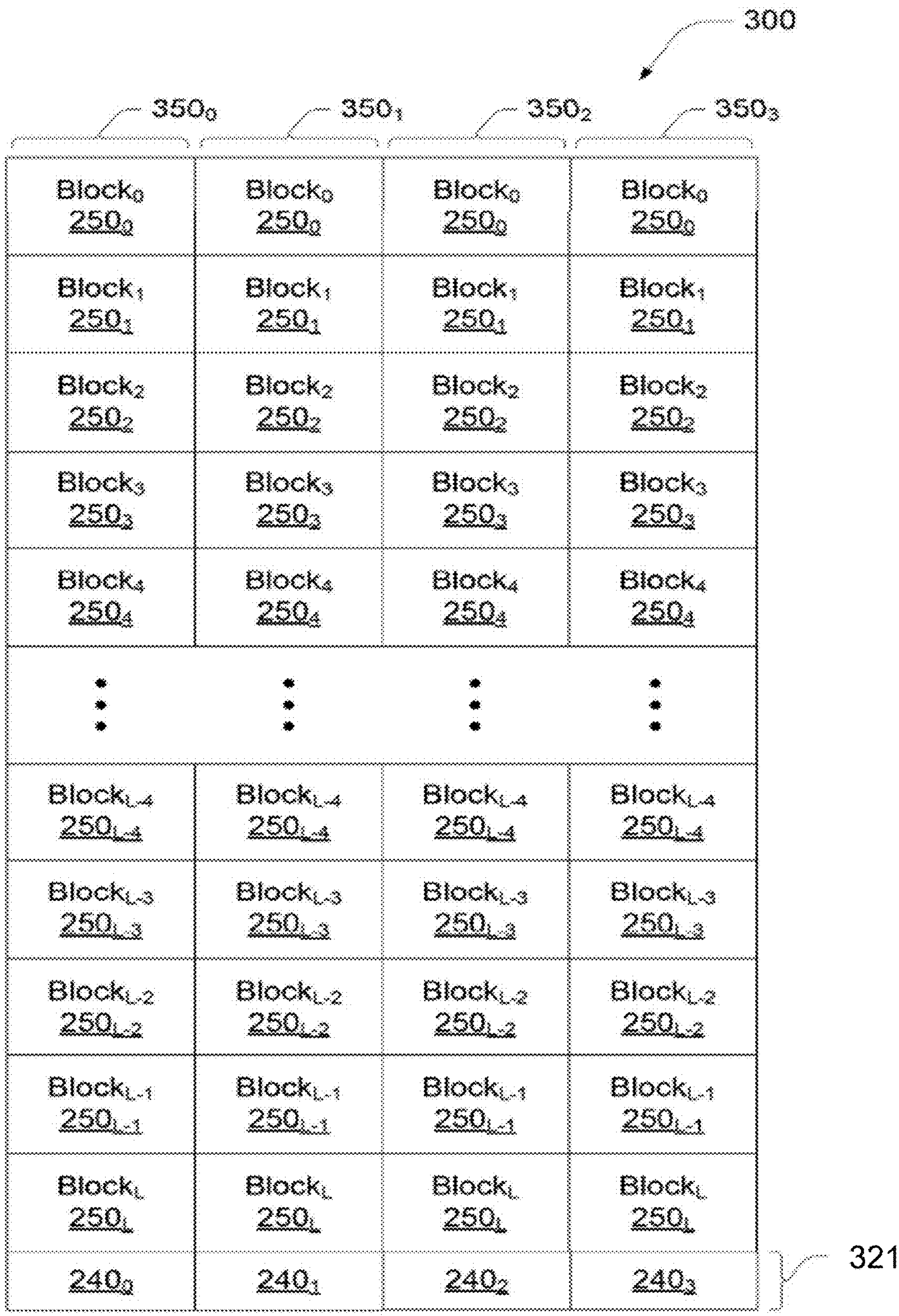
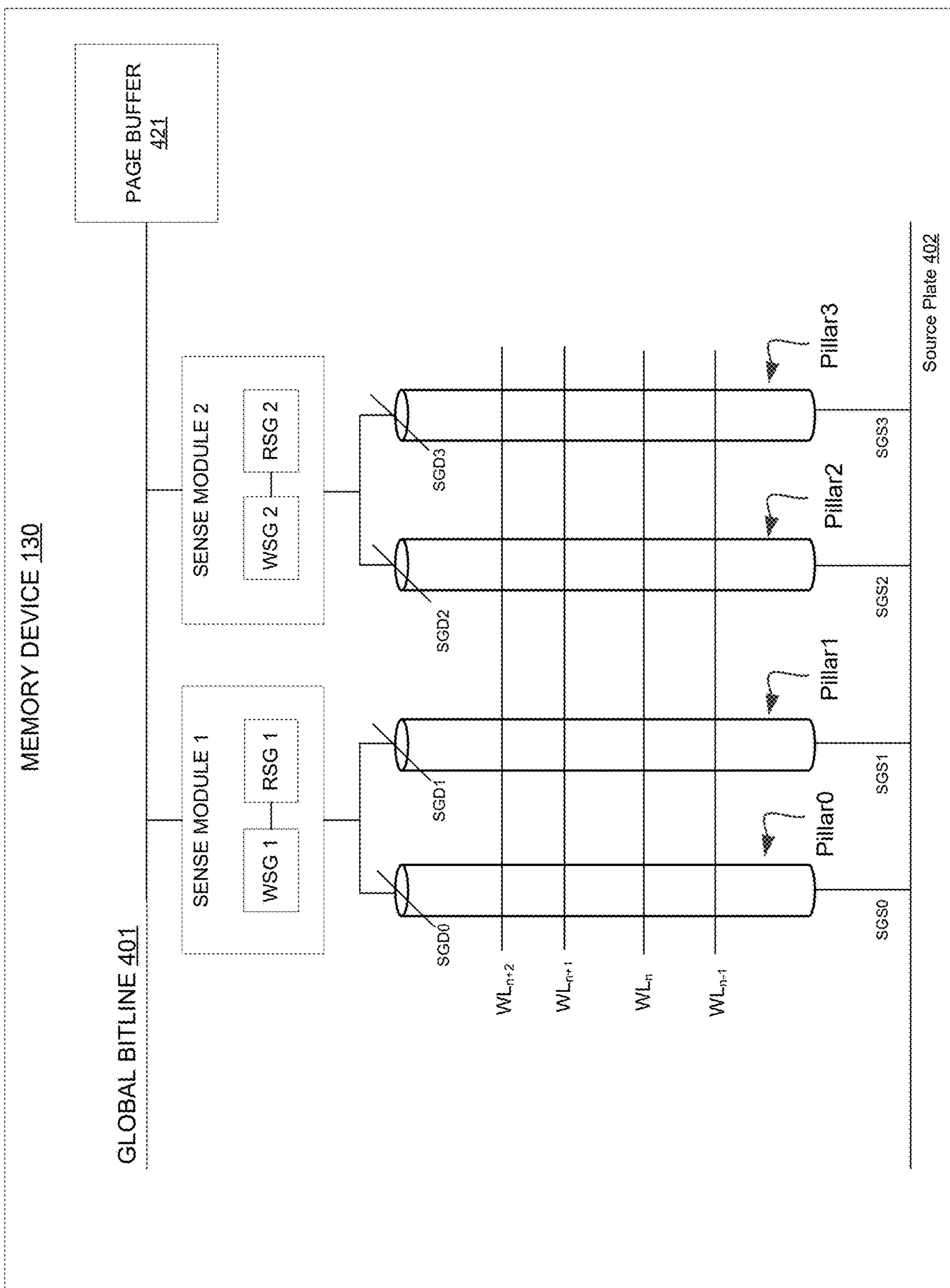

300
350_0
350_1
350_2
350_3
$Block_0$ 250_0
$Block_1$ 250_1
$Block_2$ 250_2
$Block_3$ 250_3
$Block_4$ 250_4
$Block_{L-4}$ 250_{L-4}
$Block_{L-3}$ 250_{L-3}
$Block_{L-2}$ 250_{L-2}
$Block_{L-1}$ 250_{L-1}
$Block_L$ 250_L
240_0
240_1
240_2
240_3
321

FIG. 3

FIG. 4



MEMORY DEVICE 130
PAGE BUFFER 421
GLOBAL BITLINE 401
SENSE MODULE 1
WSG 1
RSG 1
SENSE MODULE 2
WSG 2
RSG 2
SGD0
SGD1
SGD2
SGD3
Pillar0
Pillar1
Pillar2
Pillar3
WL_{n+2}
WL_{n+1}
WL_n
WL_{n-1}
SGS0
SGS1
SGS2
SGS3
Source Plate 402

FIG. 5

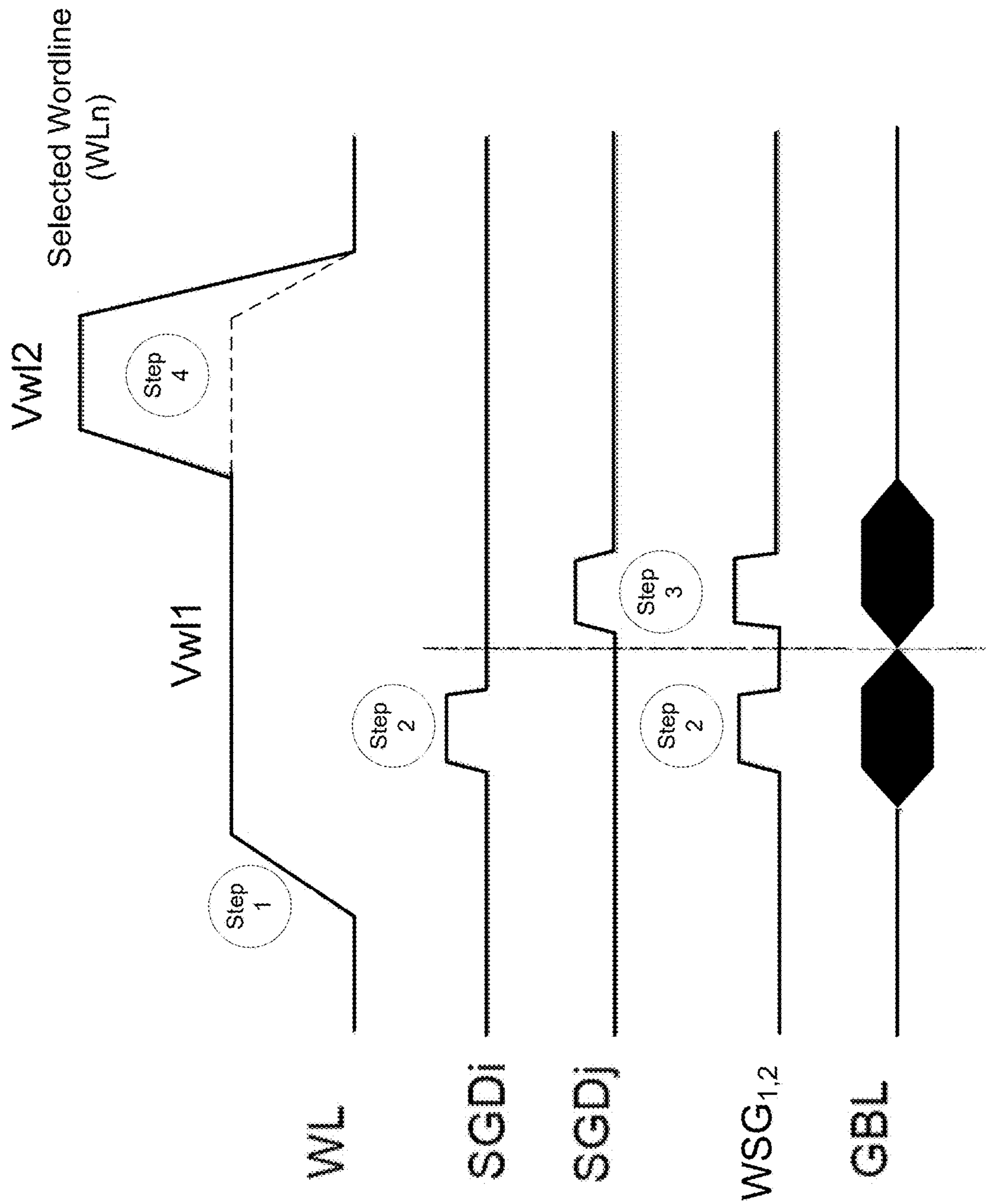

Selected Wordline
(WLn)
Vwl2
Vwl1
Step 1
Step 2
Step 2
Step 3
Step 4
WL
SGDi
SGDj
WSG1,2
GBL

FIG. 6A

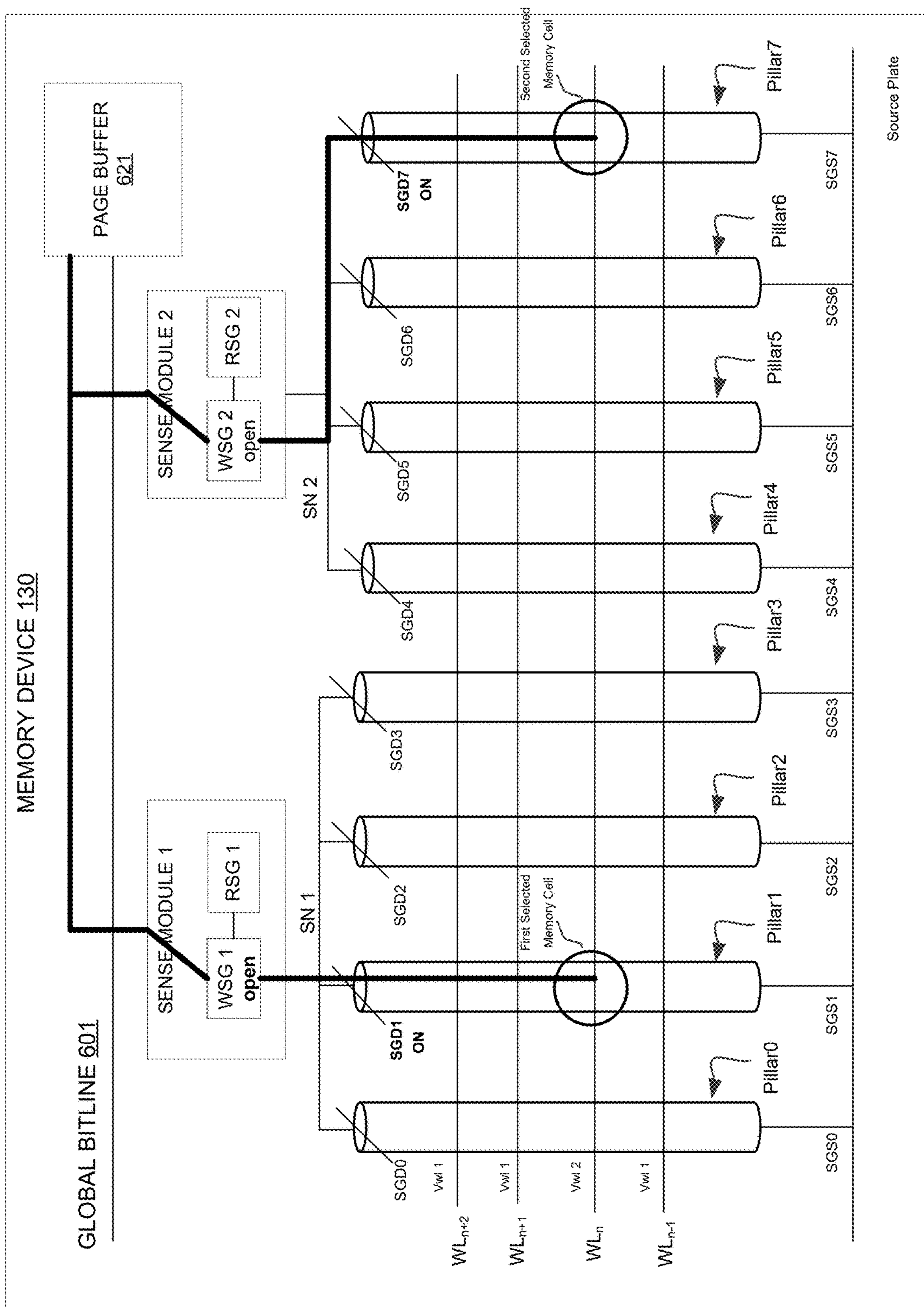

MEMORY DEVICE 130
GLOBAL BITLINE 601
PAGE BUFFER 621
SENSE MODULE 1
WSG 1 open
RSG 1
SENSE MODULE 2
WSG 2 open
RSG 2
SN 1
SN 2
SGD0
SGD1 ON
SGD2
SGD3
SGD4
SGD5
SGD6
SGD7 ON
First Selected Memory Cell
Second Selected Memory Cell
Vwl 1
Vwl 1
Vwl 2
Vwl 1
WL_{n+2}
WL_{n+1}
WL_n
WL_{n-1}
Pillar0
Pillar1
Pillar2
Pillar3
Pillar4
Pillar5
Pillar6
Pillar7
SGS0
SGS1
SGS2
SGS3
SGS4
SGS5
SGS6
SGS7
Source Plate

FIG. 6B

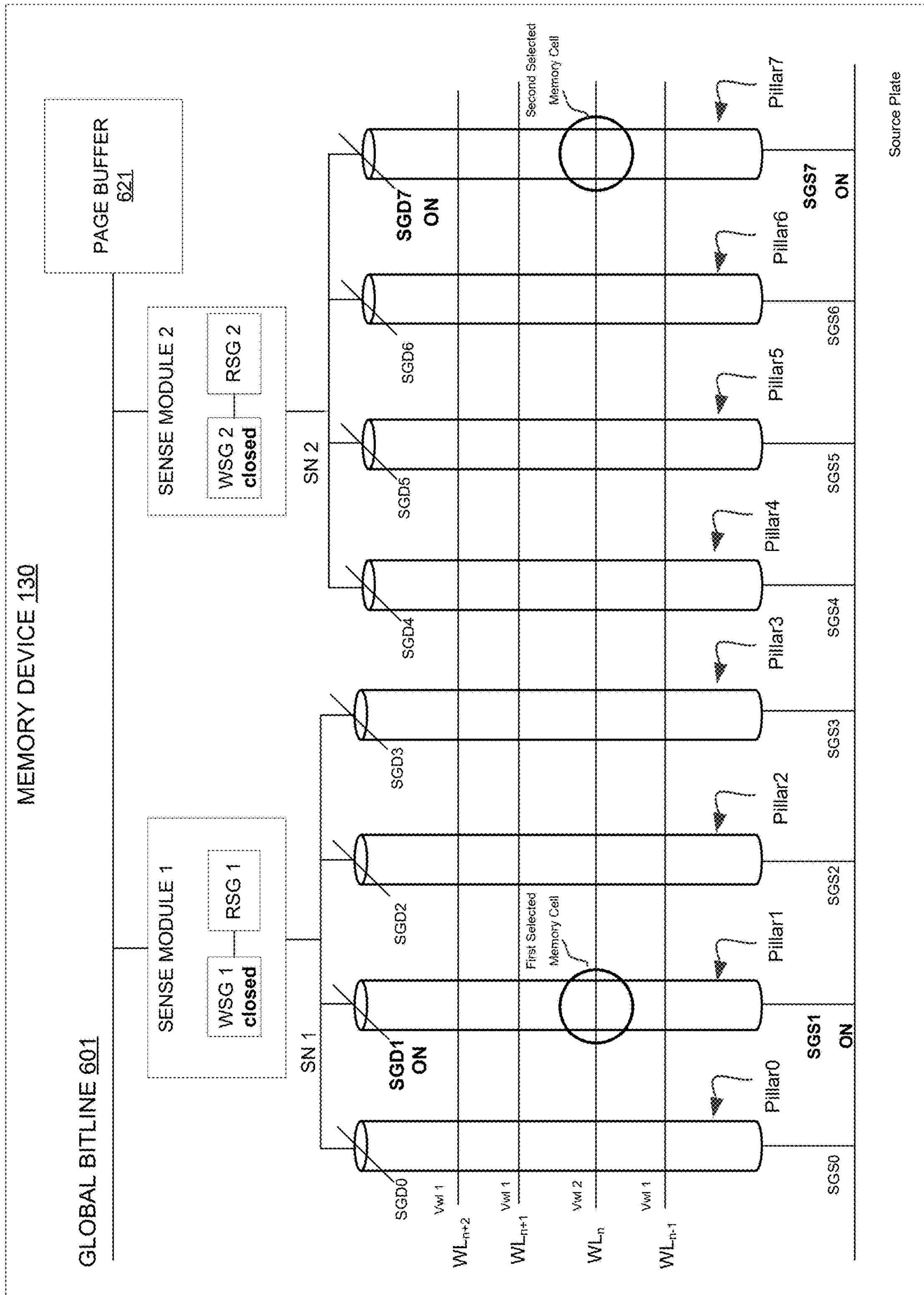

MEMORY DEVICE 130
GLOBAL BITLINE 601
PAGE BUFFER 621
SENSE MODULE 1
WSG 1 closed
RSG 1
SENSE MODULE 2
WSG 2 closed
RSG 2
SN 1
SN 2
SGD0
SGD1 ON
SGD2
SGD3
SGD4
SGD5
SGD6
SGD7 ON
WL_{n+2}
WL_{n+1}
WL_n
WL_{n-1}
Vwl 1
Vwl 1
Vwl 2
Vwl 1
First Selected Memory Cell
Second Selected Memory Cell
Pillar0
Pillar1
Pillar2
Pillar3
Pillar4
Pillar5
Pillar6
Pillar7
SGS0
SGS1 ON
SGS2
SGS3
SGS4
SGS5
SGS6
SGS7 ON
Source Plate

FIG. 6C

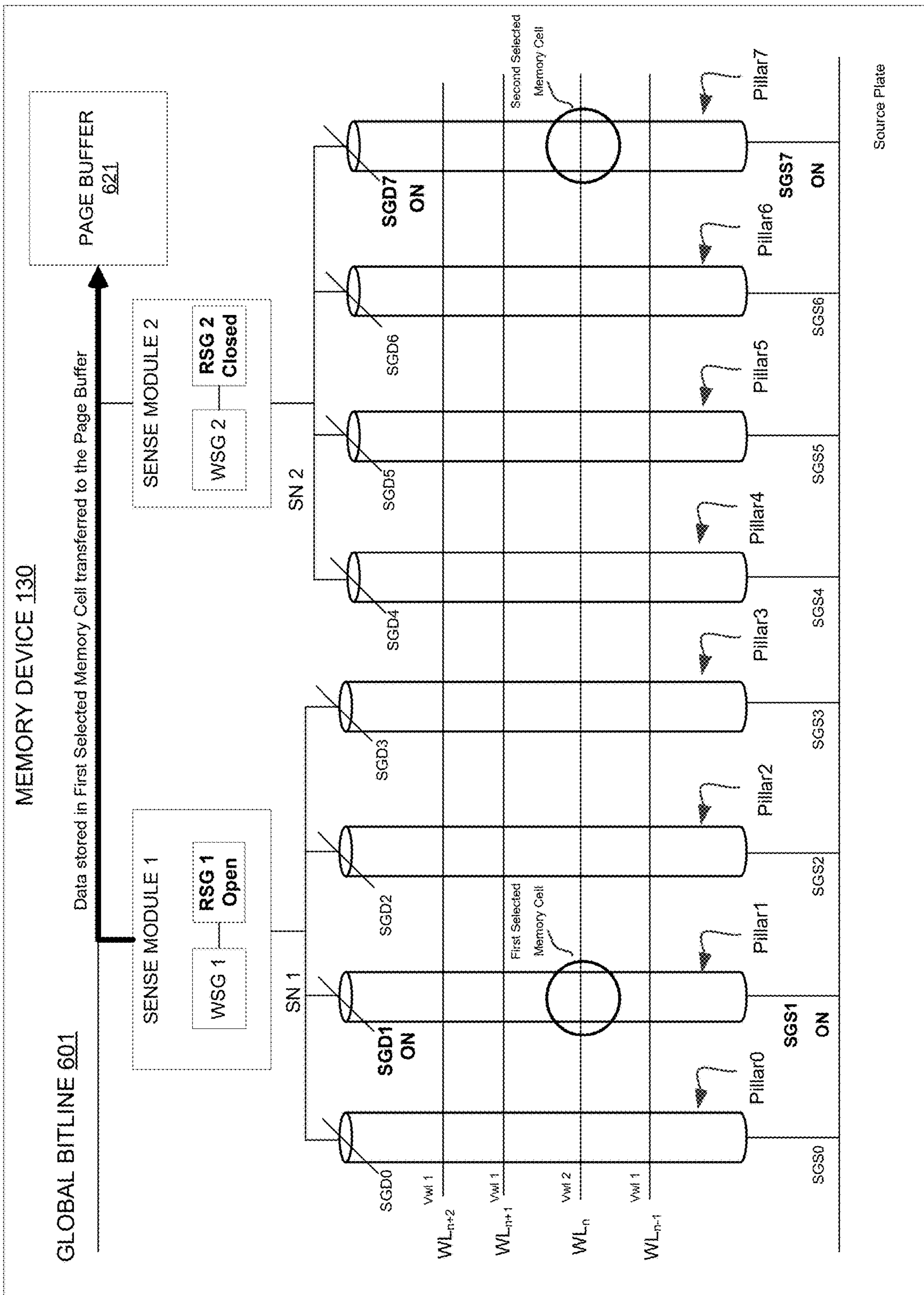

MEMORY DEVICE 130
GLOBAL BITLINE 601
Data stored in First Selected Memory Cell transferred to the Page Buffer
PAGE BUFFER
621
SENSE MODULE 1
WSG 1
RSG 1
Open
SENSE MODULE 2
WSG 2
RSG 2
Closed
SN 1
SN 2
SGD0
SGD1
ON
SGD2
SGD3
SGD4
SGD5
SGD6
SGD7
ON
WL_{n+2}
WL_{n+1}
WL_n
WL_{n-1}
Vwl 1
Vwl 1
Vwl 2
Vwl 1
First Selected Memory Cell
Second Selected Memory Cell
Pillar0
Pillar1
Pillar2
Pillar3
Pillar4
Pillar5
Pillar6
Pillar7
SGS0
SGS1
ON
SGS2
SGS3
SGS4
SGS5
SGS6
SGS7
ON
Source Plate

FIG. 6D

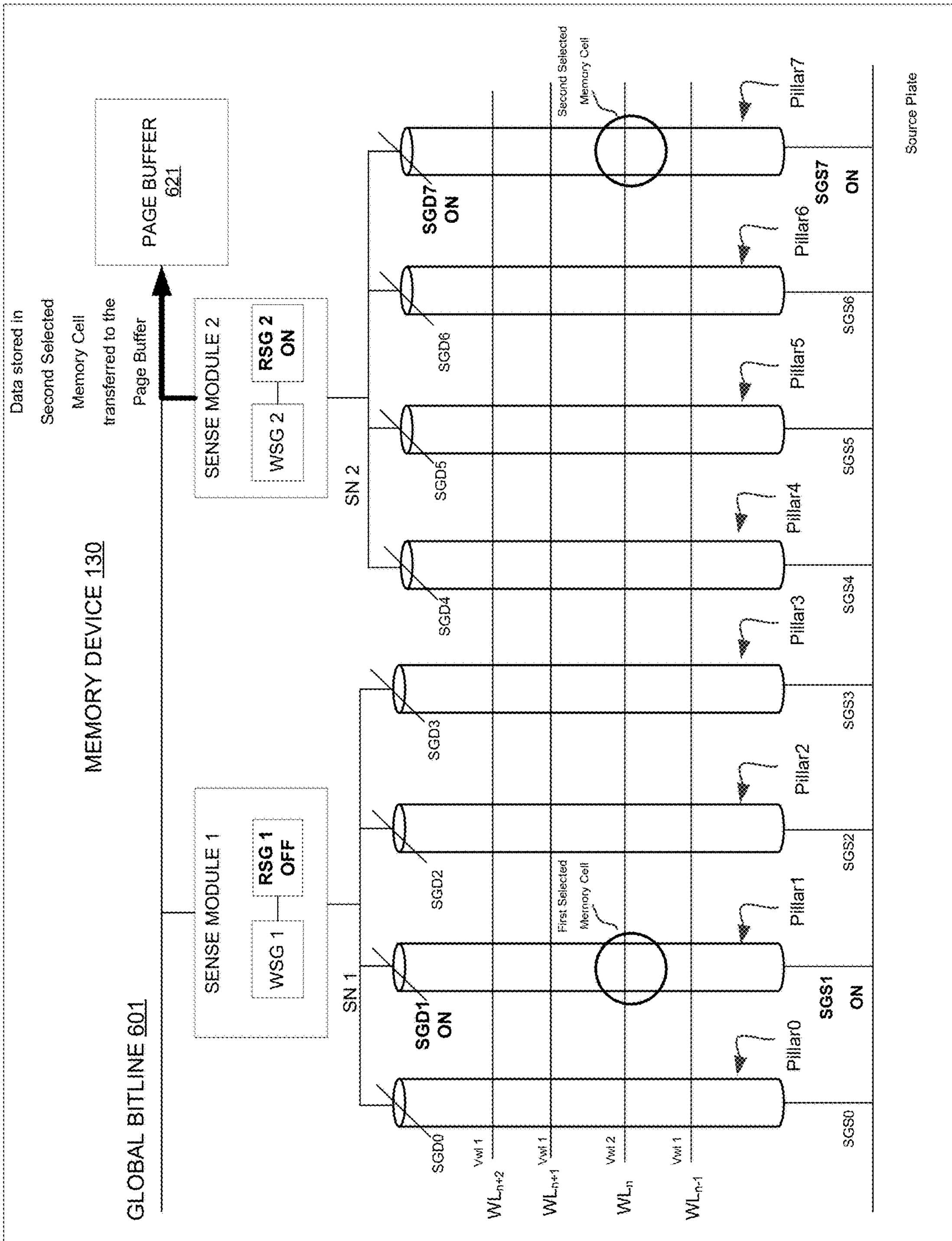

MEMORY DEVICE 130
GLOBAL BITLINE 601
PAGE BUFFER 621
Data stored in Second Selected Memory Cell transferred to the Page Buffer
SENSE MODULE 1
WSG 1
RSG 1 OFF
SENSE MODULE 2
WSG 2
RSG 2 ON
SN 1
SN 2
SGD0
SGD1 ON
SGD2
SGD3
SGD4
SGD5
SGD6
SGD7 ON
WL_{n+2}
WL_{n+1}
WL_n
WL_{n-1}
Vwl 1
Vwl 1
Vwl 2
Vwl 1
First Selected Memory Cell
Second Selected Memory Cell
Pillar0
Pillar1
Pillar2
Pillar3
Pillar4
Pillar5
Pillar6
Pillar7
SGS0
SGS1 ON
SGS2
SGS3
SGS4
SGS5
SGS6
SGS7 ON
Source Plate

FIG. 7

700

Initiate a memory access operation associated with a memory device including a memory block including a first sub-block and a second sub-block, where a first sense module is coupled to the first sub-block and a second sense module is coupled to a second sub-block
710

During the memory access operation, cause transmission, via the first sense module, of first data between a page buffer circuit and a first memory cell of the first sub-block
720

During the memory access operation, cause transmission, via the second sense module, of second data between the page buffer circuit and a second memory cell of the second sub-block
730

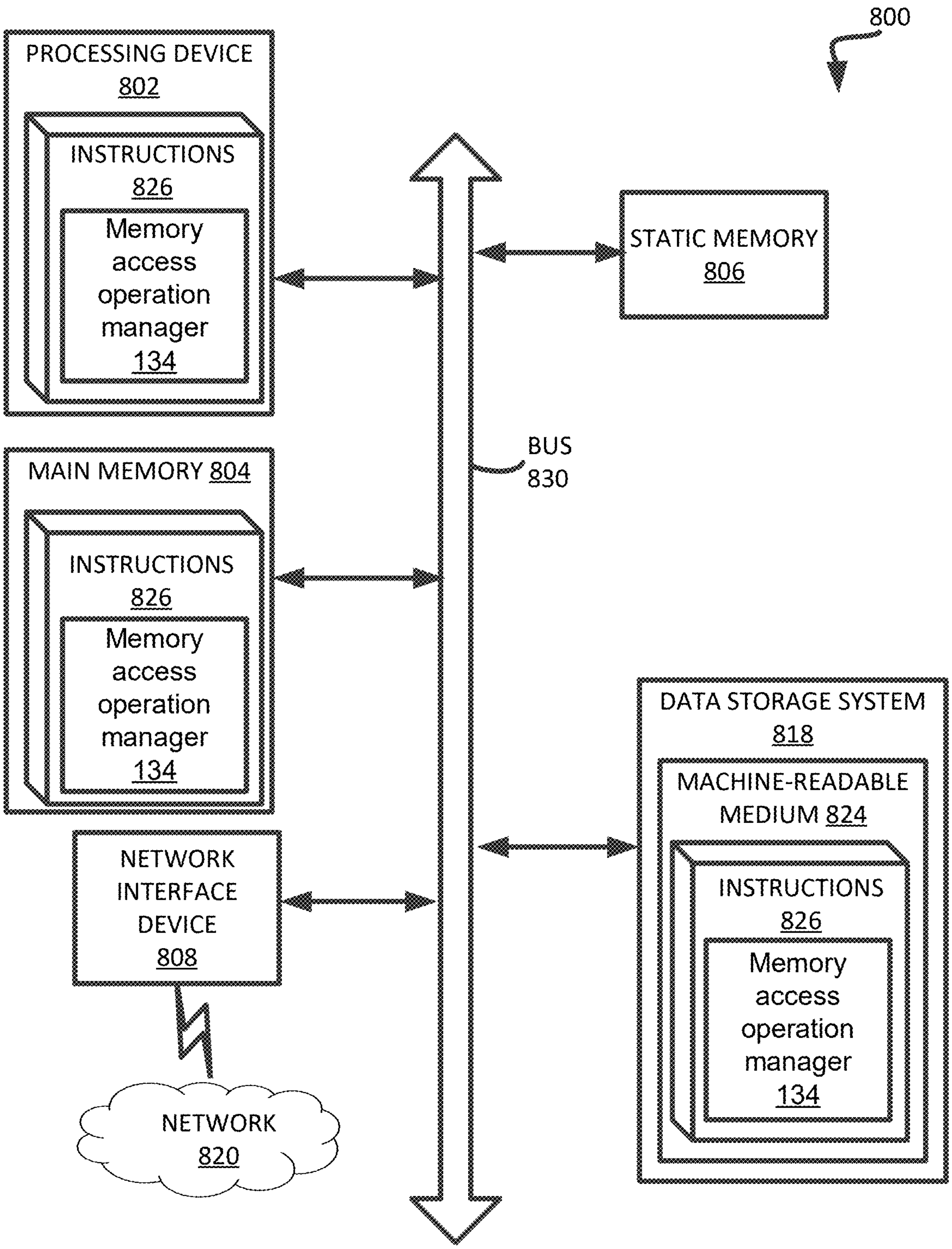

800
PROCESSING DEVICE
802
INSTRUCTIONS
826
Memory
access
operation
manager
134
STATIC MEMORY
806
BUS
830
MAIN MEMORY 804
INSTRUCTIONS
826
Memory
access
operation
manager
134
DATA STORAGE SYSTEM
818
MACHINE-READABLE
MEDIUM 824
INSTRUCTIONS
826
Memory
access
operation
manager
134
NETWORK
INTERFACE
DEVICE
808
NETWORK
820

FIG. 8

EXECUTION OF A MEMORY ACCESS OPERATION INVOLVING MULTIPLE MEMORY CELLS OF A MEMORY DEVICE USING MULTIPLE SENSE MODULES

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of U.S. Provisional Application No. 63/558,996, titled "Execution of a Memory Access Operation Involving Multiple Sub-blocks of a Memory Device Using Multiple Sense Modules," filed Feb. 28, 2024, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] Embodiments of the disclosure relate generally to memory sub-systems, and more specifically, relate to execution of a memory access operation involving multiple sub-blocks of a memory device using multiple sense modules.

BACKGROUND

[0003] A memory sub-system can include one or more memory devices that store data. The memory devices can be, for example, non-volatile memory devices and volatile memory devices. In general, a host system can utilize a memory sub-system to store data at the memory devices and to retrieve data from the memory devices.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure will be understood more fully from the detailed description given below and from the accompanying drawings of various embodiments of the disclosure.

[0005] FIG. **1**A illustrates an example computing system that includes a memory sub-system, in accordance with one or more embodiments of the present disclosure.

[0006] FIG. **1**B is a block diagram of a memory device in communication with a memory sub-system controller of a memory sub-system, in accordance with one or more embodiments of the present disclosure.

[0007] FIG. **2**A-**2**C are schematics of portions of an array of memory cells as could be used in a memory of the type described with reference to FIG. **1**B, in accordance with one or more embodiments of the present disclosure.

[0008] FIG. **3** is a block schematic of a portion of an array of memory cells as could be used in a memory of the type described with reference to FIG. **1**B, in accordance with one or more embodiments of the present disclosure.

[0009] FIG. **4** illustrates portions of an example memory device configured to execute a ganged memory access operation associated with multiple selected sub-blocks of the memory device, in accordance with one or more embodiments of the present disclosure.

[0010] FIG. **5** illustrates example waveforms corresponding to steps of a ganged program operation, in accordance with one or more embodiments of the present disclosure.

[0011] FIGS. **6**A-**6**D illustrates portions of an example memory device during execution of a ganged read or ganged program verify operation, in accordance with one or more embodiments of the present disclosure.

[0012] FIG. **7** is a flow diagram of an example method of executing a ganged media access operation, in accordance with one or more embodiments of the present disclosure.

[0013] FIG. **8** is a block diagram of an example computer system in which embodiments of the present disclosure can operate.

DETAILED DESCRIPTION

[0014] Aspects of the present disclosure are directed to execution of a memory access operation involving multiple sub-blocks of a memory device using multiple sense modules. A memory sub-system can be a storage device, a memory module, or a hybrid of a storage device and memory module. Examples of storage devices and memory modules are described below in conjunction with FIG. **1**A. In general, a host system can utilize a memory sub-system that includes one or more components, such as memory devices that store data. The host system can provide data to be stored at the memory sub-system and can request data to be retrieved from the memory sub-system.

[0015] A memory sub-system can include high density non-volatile memory devices where retention of data is desired when no power is supplied to the memory device. One example of non-volatile memory devices is a negative-and (NAND) memory device. Other examples of non-volatile memory devices are described below in conjunction with FIG. **1**A. A non-volatile memory device is a package of one or more dies. Each die can consist of one or more planes. For some types of non-volatile memory devices (e.g., NAND devices), each plane consists of a set of physical blocks. Each block consists of a set of pages. Each page consists of a set of memory cells ("cells"). A cell is an electronic circuit that stores information. Depending on the cell type, a cell can store one or more bits of binary information, and has various logic states that correlate to the number of bits being stored. The logic states can be represented by binary values, such as "0" and "1", or combinations of such values.

[0016] Memory cells are formed on a silicon wafer in an array of columns (also hereinafter referred to as "bitlines") and rows (also hereinafter referred to as wordlines). A wordline can refer to one or more rows of memory cells of a memory device that are used with one or more bitlines to generate the address of each of the memory cells. The intersection of a bitline and wordline constitutes the address of the memory cell.

[0017] A block hereinafter refers to a unit of the memory device used to store data and can include a group of memory cells, a wordline group, a wordline, or individual memory cells. Each block can include a number of sub-blocks, where each sub-block is defined by an associated pillar (e.g., a vertical conductive trace) extending from a shared bitline. Memory pages (also referred to herein as "pages") store one or more bits of binary data corresponding to data received from the host system. To achieve high density, a string of memory cells in a non-volatile memory device can be constructed to include a number of memory cells at least partially surrounding a pillar of poly-silicon channel material (i.e., a channel region). The memory cells can be coupled to access lines (i.e., wordlines) often fabricated in common with the memory cells, so as to form an array of strings in a block of memory (e.g., a memory array). The compact nature of certain non-volatile memory devices, such as 3D flash NAND memory, means wordlines are common to many memory cells within a block of memory. Some memory devices use certain types of memory cells, such as triple-level cell (TLC) memory cells, which store three bits of data in each memory cell, which make it affordable to move more applications from legacy hard disk drives to newer memory sub-systems, such as NAND solid-state drives (SSDs).

[0018] Memory access operations (e.g., a read operation, a program operation, an erase operation, etc.) can be executed with respect to the memory cells by applying a wordline bias voltage to wordlines to which memory cells of a selected page are connected. For example, during a programming operation, one or more selected memory cells can be programmed with the application of a programming voltage to a selected wordline. In one approach, an Incremental Step Pulse Programming (ISPP) process or scheme can be employed to maintain a tight cell threshold voltage distribution for higher data reliability. In ISPP, a series of high-amplitude pulses of voltage levels having an increasing magnitude (e.g., where the magnitude of subsequent pulses are increased by a predefined pulse step height) are applied to wordlines to which one or more memory cells are connected to gradually raise the voltage level of the memory cells to above a wordline voltage level corresponding to the memory access operation (e.g., a target program level). The application of the uniformly increasing pulses by a wordline driver of the memory device enables the selected wordline to be ramped or increased to a wordline voltage level (Vwl) corresponding to a memory access operation. Similarly, a series of voltage pulses having a uniformly increasing voltage level can be applied to the wordline to ramp the wordline to the corresponding wordline voltage level during the execution of an erase operation.

[0019] The series of incrementing voltage programming pulses are applied to the selected wordline to increase a charge level, and thereby a threshold voltage, of each memory cell connected to that wordline. After each programming pulse, or after a number of programming pulses, a program verify operation is performed to determine if the threshold voltage of the one or more memory cells has increased to a desired programming level (e.g., a stored target threshold voltage corresponding to a programming level). A program verify operation can include storing a target threshold voltage in a page buffer (or page buffer circuit) that is coupled to each data line (e.g., bitline) and applying a ramped voltage to the control gate of the memory cell being verified. When the ramped voltage reaches the threshold voltage to which the memory cell has been programmed, the memory cell turns on and sense circuitry detects a current on a bit line coupled to the memory cell. The detected current activates the sense circuitry to compare if the present threshold voltage is greater than or equal to the stored target threshold voltage. If the present threshold voltage is greater than or equal to the target threshold voltage, further programming is inhibited.

[0020] During programming, each pulse in the sequence of programming pulses can be incrementally increased in value (e.g., by a step voltage value such as 0.33V) to increase a charge stored on a charge storage structure corresponding to each pulse. The memory device can reach a target programming level voltage for a particular programming level by incrementally storing or increasing amounts of charge corresponding to the programming step voltage.

[0021] As the size of the memory device increases, a larger number of wordline tiers are stacked and the number of pages per block and block size increases exponentially. As a result, garbage collection operations associated with the larger blocks takes additional time to execute, which leads to a deterioration of the quality of service (QOS) metrics associated with the memory device.

[0022] According to aspects of the present disclosure, memory access operations (e.g., a program operation, a read operation, a program-verify operation) can be performed on multiple memory sub-blocks (i.e., multiple pillars) concurrently (i.e., as part of the same memory access operation). According to embodiments, multiple sense modules or circuits (herein "sense modules") are placed in connection with a set of pillars of the memory device string, where each pillar corresponds to a different sub-block of the memory device. In an embodiment, a first sense module is coupled to a first subset of sub-blocks or pillars (e.g., a first subset of two pillars, a first subset of four pillars, etc.) and a second sense module is coupled to a second subset of sub-blocks or pillars (e.g., a second subset of two pillars, a second subset of four pillars, etc.). Accordingly, the respective subsets of pillars are grouped with respect to a corresponding sense module (e.g., a first subset of sub-blocks are grouped in connection with a first sense module and a second subset of sub-blocks are grouped in connection with a second sense module).

[0023] The arrangement of multiple sense modules, each coupled to a different subset of sub-blocks enables multiple pages to be selected within a block and read or programmed concurrently (e.g., execution of a ganged read operation, a ganged program verify operation, or a ganged program operation). Advantageously, the time associated with execution of ganged read, ganged program verify, and ganged program operations relating to the multiple selected pages is reduced.

[0024] FIG. **1**A illustrates an example computing system **100** that includes a memory sub-system **110** in accordance with some embodiments of the present disclosure. The memory sub-system **110** can include media, such as one or more volatile memory devices (e.g., memory device **140**), one or more non-volatile memory devices (e.g., memory device **130**), or a combination of such.

[0025] A memory sub-system **110** can be a storage device, a memory module, or a hybrid of a storage device and memory module. Examples of a storage device include a solid-state drive (SSD), a flash drive, a universal serial bus (USB) flash drive, an embedded Multi-Media Controller (eMMC) drive, a Universal Flash Storage (UFS) drive, a secure digital (SD) and a hard disk drive (HDD). Examples of memory modules include a dual in-line memory module (DIMM), a small outline DIMM (SO-DIMM), and various types of non-volatile dual in-line memory module (NVDIMM).

[0026] The computing system **100** can be a computing device such as a desktop computer, laptop computer, network server, mobile device, a vehicle (e.g., airplane, drone, train, automobile, or other conveyance), Internet of Things (IoT) enabled device, embedded computer (e.g., one included in a vehicle, industrial equipment, or a networked commercial device), or such computing device that includes memory and a processing device.

[0027] The computing system **100** can include a host system **120** that is coupled to one or more memory sub-systems **110**. In some embodiments, the host system **120** is coupled to different types of memory sub-system **110**. FIG. **1**A illustrates one example of a host system **120** coupled to one memory sub-system **110**. As used herein, "coupled to" or "coupled with" generally refers to a connection between components, which can be an indirect communicative connection or direct communicative connection (e.g., without intervening components), whether wired or wireless, including connections such as electrical, optical, magnetic, etc.

[0028] The host system **120** can include a processor chipset and a software stack executed by the processor chipset. The processor chipset can include one or more cores, one or more caches, a memory controller (e.g., NVDIMM controller), and a storage protocol controller (e.g., PCIe controller, SATA controller, compute express link (CXL) interface). The host system **120** uses the memory sub-system **110**, for example, to write data to the memory sub-system **110** and read data from the memory sub-system **110**.

[0029] The host system **120** can be coupled to the memory sub-system **110** via a physical host interface. Examples of a physical host interface include, but are not limited to, a serial advanced technology attachment (SATA) interface, a CXL interface, a peripheral component interconnect express (PCIe) interface, universal serial bus (USB) interface, Fibre Channel, Serial Attached SCSI (SAS), a double data rate (DDR) memory bus, Small Computer System Interface (SCSI), a dual in-line memory module (DIMM) interface (e.g., DIMM socket interface that supports Double Data Rate (DDR)), etc. The physical host interface can be used to transmit data between the host system **120** and the memory sub-system **110**. The host system **120** can further utilize an NVM Express (NVMe) interface to access components (e.g., memory devices **130**) when the memory sub-system **110** is coupled with the host system **120** by the physical host interface (e.g., PCIe or CXL bus). The physical host interface can provide an interface for passing control, address, data, and other signals between the memory sub-system **110** and the host system **120**. FIG. **1**A illustrates a memory sub-system **110** as an example. In general, the host system **120** can access multiple memory sub-systems via a same communication connection, multiple separate communication connections, and/or a combination of communication connections.

[0030] The memory devices **130**,**140** can include any combination of the different types of non-volatile memory devices and/or volatile memory devices. The volatile memory devices (e.g., memory device **140**) can be, but are not limited to, random access memory (RAM), such as dynamic random access memory (DRAM) and synchronous dynamic random access memory (SDRAM).

[0031] Some examples of non-volatile memory devices (e.g., memory device **130**) include negative-and (NAND) type flash memory and write-in-place memory, such as a three-dimensional cross-point ("3D cross-point") memory device, which is a cross-point array of non-volatile memory cells. A cross-point array of non-volatile memory can perform bit storage based on a change of bulk resistance, in conjunction with a stackable cross-gridded data access array. Additionally, in contrast to many flash-based memories, cross-point non-volatile memory can perform a write in-place operation, where a non-volatile memory cell can be programmed without the non-volatile memory cell being previously erased. NAND type flash memory includes, for example, two-dimensional NAND (2D NAND) and three-dimensional NAND (3D NAND).

[0032] Each of the memory devices **130** can include one or more arrays of memory cells. One type of memory cell, for example, single level cells (SLC) can store one bit per cell. Other types of memory cells, such as multi-level cells (MLCs), triple level cells (TLCs), quad-level cells (QLCs), and penta-level cells (PLCs) can store multiple bits per cell. In some embodiments, each of the memory devices **130** can include one or more arrays of memory cells such as SLCs, MLCs, TLCs, QLCs, or any combination of such. In some embodiments, a particular memory device can include an SLC portion, and an MLC portion, a TLC portion, a QLC portion, or a PLC portion of memory cells. The memory cells of the memory devices **130** can be grouped as pages that can refer to a logical unit of the memory device used to store data. With some types of memory (e.g., NAND), pages can be grouped to form blocks. In one embodiment, the term "MLC memory" can be used to represent any type of memory cell that stores more than one bit per cell (e.g., 2 bits, 3 bits, 4 bits, or 5 bits per cell).

[0033] Although non-volatile memory components such as 3D cross-point array of non-volatile memory cells and NAND type flash memory (e.g., 2D NAND, 3D NAND) are described, the memory device **130** can be based on any other type of non-volatile memory, such as read-only memory (ROM), phase change memory (PCM), self-selecting memory, other chalcogenide based memories, ferroelectric transistor random-access memory (FeTRAM), ferroelectric random access memory (FeRAM), magneto random access memory (MRAM), Spin Transfer Torque (STT)-MRAM, conductive bridging RAM (CBRAM), resistive random access memory (RRAM), oxide based RRAM (OxRAM), negative-or (NOR) flash memory, and electrically erasable programmable read-only memory (EEPROM).

[0034] A memory sub-system controller **115** (or controller **115** for simplicity) can communicate with the memory devices **130** to perform operations such as reading data, writing data, or erasing data at the memory devices **130** and other such operations. The memory sub-system controller **115** can include hardware such as one or more integrated circuits and/or discrete components, a buffer memory, or a combination thereof. The hardware can include a digital circuitry with dedicated (i.e., hard-coded) logic to perform the operations described herein. The memory sub-system controller **115** can be a microcontroller, special purpose logic circuitry (e.g., a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), etc.), or other suitable processor.

[0035] The memory sub-system controller **115** can be a processing device, which includes one or more processors (e.g., processor **117**), configured to execute instructions stored in a local memory **119**. In the illustrated example, the local memory **119** of the memory sub-system controller **115** includes an embedded memory configured to store instructions for performing various processes, operations, logic flows, and routines that control operation of the memory sub-system **110**, including handling communications between the memory sub-system **110** and the host system **120**.

[0036] In some embodiments, the local memory **119** can include memory registers storing memory pointers, fetched data, etc. The local memory **119** can also include read-only memory (ROM) for storing micro-code. While the example memory sub-system **110** in FIG. **1**A has been illustrated as including the memory sub-system controller **115**, in another embodiment of the present disclosure, a memory sub-system **110** does not include a memory sub-system controller **115**, and can instead rely upon external control (e.g., provided by an external host, or by a processor or controller separate from the memory sub-system).

[0037] In general, the memory sub-system controller **115** can receive commands or operations from the host system **120** and can convert the commands or operations into instructions or appropriate commands to achieve the desired access to the memory devices **130**. The memory sub-system controller **115** can be responsible for other operations such as wear leveling operations, garbage collection operations, error detection and error-correcting code (ECC) operations, encryption operations, caching operations, and address translations between a logical address (e.g., logical block address (LBA), namespace) and a physical address (e.g., physical block address) that are associated with the memory devices **130**. The memory sub-system controller **115** can further include host interface circuitry to communicate with the host system **120** via the physical host interface. The host interface circuitry can convert the commands received from the host system into command instructions to access the memory devices **130** as well as convert responses associated with the memory devices **130** into information for the host system **120**.

[0038] The memory sub-system **110** can also include additional circuitry or components that are not illustrated. In some embodiments, the memory sub-system **110** can include a cache or buffer (e.g., DRAM) and address circuitry (e.g., a row decoder and a column decoder) that can receive an address from the memory sub-system controller **115** and decode the address to access the memory devices **130**.

[0039] In some embodiments, the memory devices **130** include local media controllers **135** that operate in conjunction with memory sub-system controller **115** to execute operations on one or more memory cells of the memory devices **130**. An external controller (e.g., memory sub-system controller **115**) can externally manage the memory device **130** (e.g., perform media management operations on the memory device **130**). In some embodiments, memory sub-system **110** is a managed memory device, which includes a raw memory device **130** having control logic (e.g., local media controller **135**) on the die and a controller (e.g., memory sub-system controller **115**) for media management within the same memory device package. An example of a managed memory device is a managed NAND (MNAND) device.

[0040] In one embodiment, the memory sub-system **110** includes a memory interface component **113**. Memory interface component **113** is responsible for handling interactions of memory sub-system controller **115** with the memory devices of memory sub-system **110**, such as memory device **130**. For example, memory interface component **113** can send memory access commands corresponding to requests received from host system **120** to memory device **130**, such as program commands, read commands, or other commands. In addition, memory interface component **113** can receive data from memory device **130**, such as data retrieved in response to a read command or a confirmation that a program command was successfully performed. For example, the memory sub-system controller **115** can include a processor **117** (processing device) configured to execute instructions stored in local memory **119** for performing the operations described herein.

[0041] In one embodiment, memory device **130** includes a memory access operation manager **134** configured to execute memory access operations associated with multiple selected pages in a block of the memory device **130**. In some embodiments, local media controller **135** includes at least a portion of memory access operation manager **134** and is configured to perform the functionality described herein. In some embodiments, memory access operation manager **134** is implemented on memory device **130** using firmware, hardware components, or a combination of the above. In one embodiment, memory access operation manager **134** receives, from a requestor, such as memory interface **113**, a command to execute a read operation, a program verify operation, or a program operation associated with multiple selected pages (also referred to as a "ganged read operation", a "ganged program verify operation", or a "ganged program operation") associated with a memory array of memory device **130**. The memory array can include an array of memory cells formed at the intersections of wordlines and bitlines. In one embodiment, the memory cells are grouped in to blocks, which can be further divided into sub-blocks (i.e., pillars), where a given wordline is shared across a number of sub-blocks, for example. In one embodiment, each sub-block corresponds to a separate plane in the memory array. The group of memory cells associated with a wordline within a sub-block is referred to as a physical page. In one embodiment, there can be multiple portions of the memory array, such as a first portion where the sub-blocks are configured as SLC memory and a second portion where the sub-blocks are configured as multi-level cell (MLC) memory (i.e., including memory cells that can store two or more bits of information per cell). For example, the second portion of the memory array can be configured as TLC memory. The voltage levels of the memory cells in TLC memory form a set of 8 programming distributions representing the 8 different combinations of the three bits stored in each memory cell. Depending on how the memory cells are configured, each physical page in one of the sub-blocks can include multiple page types. For example, a physical page formed from single level cells (SLCs) has a single page type referred to as a lower logical page (LP). Multi-level cell (MLC) physical page types can include LPs and upper logical pages (UPs), TLC physical page types are LPs, UPs, and extra logical pages (XPs), and QLC physical page types are LPs, UPs, XPs and top logical pages (TPs). For example, a physical page formed from memory cells of the QLC memory type can have a total of four logical pages, where each logical page can store data distinct from the data stored in the other logical pages associated with that physical page.

[0042] In one embodiment, memory access operation manager **134** causes execution of the ganged memory access operation relating to multiple pages of a block of the memory device **130**. According to embodiments, the memory access operation manager **134** can manage multiple "local" sense modules connected to a global bitline, where the multiple sense modules are placed on a same memory block. According to embodiments, each of the multiple sense modules are coupled to a subset of sub-blocks (e.g., a subset of two pillars, a subset of four pillars), where each pillar is associated with a different sub-block. Advantageously, the memory access operation manager **134** causes execution of a ganged operation (e.g., a ganged read operation, a ganged program verify operation, and a ganged program operation) where multiple pages can be selected within a block and read or programmed concurrently (i.e., as part of the same memory access operation).

[0043] According to embodiments, the global bitline is commonly connected to the set of sense modules. In operation, data can be transferred to a page buffer circuit (i.e., as part of a program operation) or transferred from the page buffer circuit (i.e., as part of a read operation or program verify operation). For example, the memory access operation manager **134** causes execution of a ganged program operation where, at a first time, first program data is provided by the page buffer circuit to a first sense module coupled to a first subset of sub-blocks including a first selected or target memory cell. In this example, at a second time, second program data is provided by the page buffer circuit to a second sense module coupled to a second subset of sub-blocks including a second selected or target memory cell.

[0044] In another example, the memory access operation manager **134** causes execution of a ganged read operation or ganged program verify operation where, at a first time, first read data is sensed by a first sense module coupled to a first subset of sub-blocks including a first selected or target memory cell and provided to the page buffer circuit. In this example, at a second time, second read data is sensed by a second sense module coupled to a second subset of sub-blocks including a second selected or target memory cell and provided to the page buffer circuit. Accordingly, the memory access operation manager **134** can cause read data to be alternately sensed and transmitted by the sense modules to the page buffer circuit. Further details with regards to the operations of memory access operation manager **134** are described below.

[0045] FIG. **1**B is a simplified block diagram of a first apparatus, in the form of a memory device **130**, in communication with a second apparatus, in the form of a memory sub-system controller **115** of a memory sub-system (e.g., memory sub-system **110** of FIG. **1**A), according to an embodiment. Some examples of electronic systems include personal computers, personal digital assistants (PDAs), digital cameras, digital media players, digital recorders, games, appliances, vehicles, wireless devices, mobile telephones and the like. The memory sub-system controller **115** (e.g., a controller external to the memory device **130**), may be a memory controller or other external host device.

[0046] Memory device **130** includes an array of memory cells **150** logically arranged in rows and columns. Memory cells of a logical row are typically connected to the same access line (e.g., a wordline) while memory cells of a logical column are typically selectively connected to the same data line (e.g., a bitline). A single access line may be associated with more than one logical row of memory cells and a single data line may be associated with more than one logical column. Memory cells (not shown in FIG. **1**B) of at least a portion of array of memory cells **250** are capable of being programmed to one of at least two target data states.

[0047] Row decode circuitry **108** and column decode circuitry **111** are provided to decode address signals. Address signals are received and decoded to access the array of memory cells **150**. Memory device **130** also includes input/output (I/O) control circuitry **112** to manage input of commands, addresses and data to the memory device **130** as well as output of data and status information from the memory device **130**. An address register **114** is in communication with I/O control circuitry **112** and row decode circuitry **108** and column decode circuitry **111** to latch the address signals prior to decoding. A command register **124** is in communication with I/O control circuitry **112** and local media controller **135** to latch incoming commands.

[0048] A controller (e.g., the local media controller **135** internal to the memory device **130**) controls access to the array of memory cells **150** in response to the commands and generates status information for the external memory sub-system controller **115**, i.e., the local media controller **135** is configured to perform access operations (e.g., read operations, programming operations and/or erase operations) on the array of memory cells **150**. The local media controller **135** is in communication with row decode circuitry **108** and column decode circuitry **111** to control the row decode circuitry **108** and column decode circuitry **111** in response to the addresses. In one embodiment, local media controller **135** includes memory access operation manager **134**, which can cause execution of ganged memory access operations (i.e., memory access operations associated with multiple selected pages using a set of multiple sense modules coupled to a page buffer circuit via a global bitline, where each sense module is coupled to a subset of multiple pillars, as described herein.

[0049] The local media controller **135** is also in communication with a cache register **118**. Cache register **118** latches data, either incoming or outgoing, as directed by the local media controller **135** to temporarily store data while the array of memory cells **150** is busy writing or reading, respectively, other data. During a program operation (e.g., write operation), data may be passed from the cache register **118** to a page buffer circuit **121** for transfer to the array of memory cells **150**; then new data may be latched in the cache register **118** from the I/O control circuitry **212**. During a read operation, data may be passed from the cache register **118** to the I/O control circuitry **112** for output to the memory sub-system controller **115**; then new data may be passed from the page buffer circuit **121** to the cache register **118**. The cache register **118** and/or the page buffer circuit **121** may form (e.g., may form a portion of) a page buffer of the memory device **130**. The page buffer circuit **121** may further include sensing devices (not shown in FIG. **1**B) to sense a data state of a memory cell of the array of memory cells **150**, e.g., by sensing a state of a data line connected to that memory cell. A status register **122** may be in communication with I/O control circuitry **112** and the local memory controller **135** to latch the status information for output to the memory sub-system controller **115**.

[0050] Memory device **130** receives control signals at the memory sub-system controller **115** from the local media controller **135** over a control link **132**. For example, the control signals can include a chip enable signal CE #, a command latch enable signal CLE, an address latch enable signal ALE, a write enable signal WE #, a read enable signal RE #, and a write protect signal WP #. Additional or alternative control signals (not shown) may be further received over control link **132** depending upon the nature of the memory device **130**. In one embodiment, memory device **130** receives command signals (which represent commands), address signals (which represent addresses), and data signals (which represent data) from the memory sub-system controller **115** over a multiplexed input/output (I/O) bus **133** and outputs data to the memory sub-system controller **115** over I/O bus **133**.

[0051] For example, the commands may be received over input/output (I/O) pins [7:0] of I/O bus **133** at I/O control circuitry **112** and may then be written into command register

124. The addresses may be received over input/output (I/O) pins [7:0] of I/O bus **234** at I/O control circuitry **112** and may then be written into address register **114**. The data may be received over input/output (I/O) pins [7:0] for an 8-bit device or input/output (I/O) pins [15:0] for a 16-bit device at I/O control circuitry **112** and then may be written into cache register **118**. The data may be subsequently written into page buffer circuit **121** for programming the array of memory cells **150**.

[0052] In an embodiment, cache register **118** may be omitted, and the data may be written directly into page buffer circuit **121**. Data may also be output over input/output (I/O) pins [7:0] for an 8-bit device or input/output (I/O) pins [15:0] for a 16-bit device. Although reference may be made to I/O pins, they may include any conductive node providing for electrical connection to the memory device **130** by an external device (e.g., the memory sub-system controller **115**), such as conductive pads or conductive bumps as are commonly used.

[0053] It will be appreciated by those skilled in the art that additional circuitry and signals can be provided, and that the memory device **130** of FIG. **1**B has been simplified. It should be recognized that the functionality of the various block components described with reference to FIG. **1**B may not necessarily be segregated to distinct components or component portions of an integrated circuit device. For example, a single component or component portion of an integrated circuit device could be adapted to perform the functionality of more than one block component of FIG. **1**B. Alternatively, one or more components or component portions of an integrated circuit device could be combined to perform the functionality of a single block component of FIG. **1**B. Additionally, while specific I/O pins are described in accordance with popular conventions for receipt and output of the various signals, it is noted that other combinations or numbers of I/O pins (or other I/O node structures) may be used in the various embodiments.

[0054] FIG. **2**A-**2**C are schematics of portions of an array of memory cells **200**A, such as a NAND memory array, as could be used in a memory of the type described with reference to FIG. **1**B according to an embodiment, e.g., as a portion of the array of memory cells **104**. Memory array **200**A includes access lines, such as wordlines $\mathbf{202}_0$ to $\mathbf{202}_N$, and data lines, such as bitlines $\mathbf{204}_0$ to $\mathbf{204}_M$. The wordlines **202** can be connected to global access lines (e.g., global wordlines), not shown in FIG. **2**A, in a many-to-one relationship. For some embodiments, memory array **200**A can be formed over a semiconductor that, for example, can be conductively doped to have a conductivity type, such as a p-type conductivity, e.g., to form a p-well, or an n-type conductivity, e.g., to form an n-well.

[0055] Memory array **200**A can be arranged in rows (each corresponding to a wordline **202**) and columns (each corresponding to a bitline **204**). Each column can include a string of series-connected memory cells (e.g., non-volatile memory cells), such as one of NAND strings $\mathbf{206}_0$ to $\mathbf{206}_M$. Each NAND string **206** can be connected (e.g., selectively connected) to a common source (SRC) **216** and can include memory cells $\mathbf{208}_0$ to $\mathbf{208}_N$. The memory cells **208** can represent non-volatile memory cells for storage of data. The memory cells **208** of each NAND string **206** can be connected in series between a select gate **210** (e.g., a field-effect transistor), such as one of the select gates $\mathbf{210}_0$ to $\mathbf{210}_M$ (e.g., that can be source select transistors, commonly referred to as select gate source), and a select gate **212** (e.g., a field-effect transistor), such as one of the select gates $\mathbf{212}_0$ to $\mathbf{212}_M$ (e.g., that can be drain select transistors, commonly referred to as select gate drain). Select gates $\mathbf{210}_0$ to $\mathbf{210}_M$ can be commonly connected to a select line **214**, such as a source select line (SGS), and select gates $\mathbf{212}_0$ to $\mathbf{212}_M$ can be commonly connected to a select line **215**, such as a drain select line (SGD). Although depicted as traditional field-effect transistors, the select gates **210** and **212** can utilize a structure similar to (e.g., the same as) the memory cells **208**. The select gates **210** and **212** can represent a number of select gates connected in series, with each select gate in series configured to receive a same or independent control signal.

[0056] A source of each select gate **210** can be connected to common source **216**. The drain of each select gate **210** can be connected to a memory cell $\mathbf{208}_0$ of the corresponding NAND string **206**. For example, the drain of select gate $\mathbf{210}_0$ can be connected to memory cell $\mathbf{208}_0$ of the corresponding NAND string $\mathbf{206}_0$. Therefore, each select gate **210** can be configured to selectively connect a corresponding NAND string **206** to the common source **216**. A control gate of each select gate **210** can be connected to the select line **214**.

[0057] The drain of each select gate **212** can be connected to the bitline **204** for the corresponding NAND string **206**. For example, the drain of select gate $\mathbf{212}_0$ can be connected to the bitline $\mathbf{204}_0$ for the corresponding NAND string $\mathbf{206}_0$. The source of each select gate **212** can be connected to a memory cell $\mathbf{208}_N$ of the corresponding NAND string **206**. For example, the source of select gate $\mathbf{212}_0$ can be connected to memory cell $\mathbf{208}_N$ of the corresponding NAND string $\mathbf{206}_0$. Therefore, each select gate **212** can be configured to selectively connect a corresponding NAND string **206** to the corresponding bitline **204**. A control gate of each select gate **212** can be connected to select line **215**.

[0058] The memory array **200**A in FIG. **2**A can be a quasi-two-dimensional memory array and can have a generally planar structure, e.g., where the common source **216**, NAND strings **206** and bitlines **204** extend in substantially parallel planes. Alternatively, the memory array **200**A in FIG. **2**A can be a three-dimensional memory array, e.g., where NAND strings **206** can extend substantially perpendicular to a plane containing the common source **216** and to a plane containing the bitlines **204** that can be substantially parallel to the plane containing the common source **216**.

[0059] Typical construction of memory cells **208** includes a data-storage structure **234** (e.g., a floating gate, charge trap, and the like) that can determine a data state of the memory cell (e.g., through changes in threshold voltage), and a control gate **236**, as shown in FIG. **2**A. The data-storage structure **234** can include both conductive and dielectric structures while the control gate **236** is generally formed of one or more conductive materials. In some cases, memory cells **208** can further have a defined source/drain (e.g., source) **230** and a defined source/drain (e.g., drain) **232**. The memory cells **208** have their control gates **236** connected to (and in some cases form) a wordline **202**.

[0060] A column of the memory cells **208** can be a NAND string **206** or a number of NAND strings **206** selectively connected to a given bitline **204**. A row of the memory cells **208** can be memory cells **208** commonly connected to a given wordline **202**. A row of memory cells **208** can, but need not, include all the memory cells **208** commonly connected to a given wordline **202**. Rows of the memory cells **208** can often be divided into one or more groups of physical pages of memory cells **208**, and physical pages of the memory cells **208** often include every other memory cell **208** commonly connected to a given wordline **202**. For example, the memory cells **208** commonly connected to wordline $\mathbf{202}_N$ and selectively connected to even bitlines **204** (e.g., bitlines $\mathbf{204}_0$, $\mathbf{204}_2$, $\mathbf{204}_4$, etc.) can be one physical page of the memory cells **208** (e.g., even memory cells) while memory cells **208** commonly connected to wordline $\mathbf{202}_N$ and selectively connected to odd bitlines **204** (e.g., bitlines $\mathbf{204}_1$, $\mathbf{204}_3$, $\mathbf{204}_5$, etc.) can be another physical page of the memory cells **208** (e.g., odd memory cells).

[0061] Although bitlines $\mathbf{204}_3$-$\mathbf{204}_5$ are not explicitly depicted in FIG. **2**A, it is apparent from the figure that the bitlines **204** of the array of memory cells **200**A can be numbered consecutively from bitline **2040** to bitline $\mathbf{204}_M$. Other groupings of the memory cells **208** commonly connected to a given wordline **202** can also define a physical page of memory cells **208**. For certain memory devices, all memory cells commonly connected to a given wordline can be deemed a physical page of memory cells. The portion of a physical page of memory cells (which, in some embodiments, could still be the entire row) that is read during a single read operation or programmed during a single programming operation (e.g., an upper or lower page of memory cells) can be deemed a logical page of memory cells. A block of memory cells can include those memory cells that are configured to be erased together, such as all memory cells connected to wordlines $\mathbf{202}_0$-$\mathbf{202}_N$ (e.g., all NAND strings **206** sharing common wordlines **202**). Unless expressly distinguished, a reference to a page of memory cells herein refers to the memory cells of a logical page of memory cells. Although the example of FIG. **2**A is discussed in conjunction with NAND flash, the embodiments and concepts described herein are not limited to a particular array architecture or structure, and can include other structures (e.g., SONOS, phase change, ferroelectric, etc.) and other architectures (e.g., AND arrays, NOR arrays, etc.).

[0062] FIG. **2**B is another schematic of a portion of an array of memory cells **200**B as could be used in a memory of the type described with reference to FIG. **1**B, e.g., as a portion of the array of memory cells **104**. Like numbered elements in FIG. **2**B correspond to the description as provided with respect to FIG. **2**A. FIG. **2**B provides additional detail of one example of a three-dimensional NAND memory array structure. The three-dimensional NAND memory array **200**B can incorporate vertical structures which can include semiconductor pillars where a portion of a pillar can act as a channel region of the memory cells of NAND strings **206**. The NAND strings **206** can be each selectively connected to a bitline $\mathbf{204}_0$-$\mathbf{204}_M$ by a select transistor **212** (e.g., that can be drain select transistors, commonly referred to as select gate drain) and to a common source **216** by a select transistor **210** (e.g., that can be source select transistors, commonly referred to as select gate source). Multiple NAND strings **206** can be selectively connected to the same bitline **204**. Subsets of NAND strings **206** can be connected to their respective bitlines **204** by biasing the select lines $\mathbf{215}_0$-$\mathbf{215}_K$ to selectively activate particular select transistors **212** each between a NAND string **206** and a bitline **204**. The select transistors **210** can be activated by biasing the select line **214**. Each wordline **202** can be connected to multiple rows of memory cells of the memory array **200**B. Rows of memory cells that are commonly connected to each other by a particular wordline **202** can collectively be referred to as tiers.

[0063] FIG. **2**C is a further schematic of a portion of an array of memory cells **200**C as could be used in a memory of the type described with reference to FIG. **1**B, e.g., as a portion of the array of memory cells **104**. Like numbered elements in FIG. **2**C correspond to the description as provided with respect to FIG. **2**A. The array of memory cells **200**C can include strings of series-connected memory cells (e.g., NAND strings) **206**, access (e.g., word) lines **202**, data (e.g., bit) lines **204**, select lines **214** (e.g., source select lines), select lines **215** (e.g., drain select lines) and a source **216** as depicted in FIG. **2**A. A portion of the array of memory cells **200**A can be a portion of the array of memory cells **200**C, for example.

[0064] FIG. **2**C depicts groupings of NAND strings **206** into blocks of memory cells **250**, e.g., blocks of memory cells $\mathbf{250}_0$-$\mathbf{250}_L$. Blocks of memory cells **250** can be groupings of memory cells **208** that can be erased together in a single erase operation, sometimes referred to as erase blocks. Each block of memory cells **250** can represent those NAND strings **206** commonly associated with a single select line **215**, e.g., select line $\mathbf{215}_0$. The source **216** for the block of memory cells $\mathbf{250}_0$ can be a same source as the source **216** for the block of memory cells $\mathbf{250}_L$. For example, each block of memory cells $\mathbf{250}_0$-$\mathbf{250}_L$ can be commonly selectively connected to the source **216**. Access lines **202** and select lines **214** and **215** of one block of memory cells **250** can have no direct connection to access lines **202** and select lines **214** and **215**, respectively, of any other block of memory cells of the blocks of memory cells $\mathbf{250}_0$-$\mathbf{250}_L$.

[0065] The bitlines $\mathbf{204}_0$-$\mathbf{204}_M$ can be connected (e.g., selectively connected) to a buffer portion **240**, which can be a portion of the page buffer of the memory device **130**. The buffer portion **240** can correspond to a memory plane (e.g., the set of blocks of memory cells $\mathbf{250}_0$-$\mathbf{250}_L$). The buffer portion **240** can include sense circuits (which can include sense amplifiers) for sensing data values indicated on respective bitlines **204**.

[0066] FIG. **3** is a block schematic of a portion of an array of memory cells **300** as could be used in a memory of the type described with reference to FIG. **1**B. The array of memory cells **300** is depicted as having four memory planes **350** (e.g., memory planes $\mathbf{350}_0$-$\mathbf{350}_3$), each in communication with a respective buffer portion **240**, which can collectively form a page buffer **321**. While four memory planes **350** are depicted, other numbers of memory planes **350** can be commonly in communication with a page buffer **352**. Each memory plane **350** is depicted to include L+1 blocks of memory cells **250** (e.g., blocks of memory cells $\mathbf{250}_0$-$\mathbf{250}_L$).

[0067] FIG. **4** illustrates portions of an example memory device **130** configured to execute a ganged memory access operation associated with multiple selected pages of the memory device **130**, according to embodiments of the present disclosure. As shown in the exemplary arrangement of FIG. **4**, the memory block includes four pillars (e.g., Pillar**0**, Pillar**1**, Pillar**2**, and Pillar**3**), where each pillar corresponds to a different sub-block. As illustrated, the memory device **130** includes multiple sense modules (sense module **1** and sense module **2**) each including a write select gate (e.g., WSG **1** of sense module **1**, WSG **2** of sense module **2**) and a read select gate (e.g., RSG **1** of sense module **1**, RSG **2** of sense module **2**). In the example shown in FIG. **4**, sense module **1** is coupled to a first subset of sub-blocks (i.e., Pillar**0** and Pillar**1**) and sense module **2** is coupled to a second subset of sub-blocks (i.e., Pillar**2** and Pillar**3**). During execution of a ganged program operation, sense module **1** is configured to transfer first program data from the page buffer **421** to a first selected memory cell of a selected wordline (WLn) of the first subset of sub-blocks (i.e., Pillar**0** and Pillar**1**). During execution of the ganged program operation, sense module **2** is configured to transfer second program data from the page buffer **421** to a second selected memory cell of the selected wordline (WLn) of the second subset of sub-blocks (i.e., Pillar**2** and Pillar**3**).

[0068] According to embodiments, processing logic of the memory device **130** (i.e., memory access operation manager **134** of FIG. **1**A and **1**B) cause execution of a ganged program operation to program first data to a first selected memory cell of a first sub-block (i.e., Pillar**0** and Pillar**1**) and program second data to a second selected memory cell of a second sub-block (i.e., Pillar**2** and Pillar**3**). FIG. **5** illustrates example waveforms corresponding to the steps of the ganged program operation, as described in connection with FIG. **4**.

[0069] In an embodiment, in step **1** of the ganged program operation, the processing logic causes all wordlines (e.g., WLn−**1**, WLn, WLn+**1**, and WLn+**2** of FIG. **4**) to be raised to a first wordline voltage level (Vwl**1**) to cause the channel potentials of the pillars of memory cells to capacitively couple together. For example, the first wordline voltage level (Vwl**1**) may be approximately 10V. FIG. **5** illustrates the ramping of the wordline voltage level to Vwl**1** in step **1**.

[0070] In an embodiment, in step **2**, a potential level of global bitline **401** is set at a first bitline voltage level (Vbl**1**) (e.g., approximately 0V) or a second bitline voltage level (Vbl**2**) (e.g., approximately 3V) by the associated page buffer **421**, depending on the data to be programmed. After the potential of the global bitline **401** is modulated, the voltage applied to WSG (i.e., WSG **1** of sense module **1** and WSG **2** of sense module **2**) and one of SGDi (where i=0, 1, 2, or 3) are fired to control an associated channel potential. If the first bitline voltage (Vbl**1**) of approximately 0V is applied to the global bitline **401**, an associated channel potential is discharged and grounded. If the second bitline voltage (Vbl**2**) remains on the global bitline **401**, then an associated channel potential remains the same. FIG. **5** illustrates the increasing of the voltage applied to WSG **1** and SGDi in step **2** of the ganged program operation.

[0071] In an embodiment, in step **3**, step **2** is repeated by changing the selection of SGD (i.e., selecting SGDj, where j=not i). For example, if SGDi is SGD**0** or SGD**1** (i.e., i=0 or 1), then SGDj may be SGD**2** or SGD**3** (i.e., j=2 or 3). FIG. **5** illustrates the increasing of the voltage applied to WSG **2** and SGDj in step **3** of the ganged program operation.

[0072] In an embodiment, in step **4**, a selected wordline (WLn) is raised to a second wordline voltage level (Vwl**2**), where Vwl**2** is greater than Vwl**1**). For example, Vwl**1** may be approximately 10V and Vwl**2** may be approximately 20V. Application of the Vwl**2** to the selected wordline WLn enables the programming of data corresponding to the multiple selected memory cells of the different sub-blocks concurrently. For example, if in step **2**, SGDi=SGD**0** and SGDj=SGD**3**, then a first memory cell of Pillar**0** and a second memory cell of Pillar**3** corresponding to the selected wordline (WLn) are programmed concurrently. FIG. **5** illustrates the increasing of the wordline voltage applied to the selected wordline (WLn) from the first wordline voltage (Vwl**1**) to the second wordline voltage (Vwl**2**).

[0073] According to embodiments, the global bitline **401** is commonly connected to the set of sense modules (i.e., sense module **1** and sense module **2** of FIG. **4**). As a result, the data to be programmed is transferred from the page buffer **421** in an alternating manner. For example, first program data is transferred from the page buffer **421** to sense module **1** for programming a first selected memory cell of SGDi (where i=0 or 1) followed by the transferring of second program data from the page buffer **421** to sense module **2** for programming a second selected memory cell of SGDj (where j=2 or 3).

[0074] In an embodiment, in step **5**, the processing logic can cause verification of the data programmed to the selected memory cells by reading the states of the multiple selected memory cells. If the selected memory cells are successfully programmed with the desired data, the ganged program operation is complete. If the selected memory cells are not successfully programmed, the processing logic can repeat step **1** through step **4**.

[0075] FIGS. **6**A-**6**D illustrates execution of a ganged read operation or ganged program verify operation associated with an example memory device **130**, according to embodiments of the present disclosure. As shown in FIGS. **6**A-**6**D, the example memory device **130** includes eight pillars (e.g. Pillar**0**, Pillar**1**, Pillar**2** . . . Pillar**7**) of a memory block subject to the ganged read operation or program verify operation. As shown in FIGS. **6**A-**6**D, the memory device **130** includes a set of two sense modules (i.e., sense module **1** and sense module **2**), where each sense module is coupled to a different subset of sub-blocks. In the example shown in FIGS. **6**A-**6**D, sense module **1** is coupled to a first subset of sub-blocks including Pillar**0**, Pillar**1**, Pillar**2**, and Pillar **3** and sense module **2** is coupled to a second subset of pillars including Pillar**4**, Pillar**5**, Pillar**6**, and Pillar**7**. Advantageously, the arrangement of the sense modules coupled to the different subsets of pillars enables two different sub-blocks or pages within a block to be selected and read concurrently in a single ganged read operation (or ganged program verify operation).

[0076] FIG. **6**A illustrates an example step **1** of the ganged read operation (or ganged program verify operation). As shown in FIG. **6**A, in step **1**, a page buffer **621** supplies a first voltage (e.g., approximately 3V) to a first selected memory cell and a second selected memory cell of a selected wordline (WLn) via the global bitline **601**, WSG **1** of sense module **1** (i.e., WSG **1** is made open or turned on), WSG **2** of sense module **2** (i.e., WSG **2** is made open or turned on), sense node **1** (SN **1**), sense node (SN **2**), SGDi (where i=1) and SGDj (where j=7), while applying a first wordline voltage (Vwl**1**) to a set of unselected wordlines (e.g., WLn+**2**, WLn+**1**, and WLn−**1**) and applying a second wordline voltage (Vwl**2**) to the selected wordline (WLn). In an embodiment, the first wordline voltage (Vwl**1**) is greater than the second wordline voltage (Vwl**2**). For example, Vwl**1** may be approximately 6V and Vwl**2** may be approximately 2V.

[0077] FIG. **6**B illustrates an example step **2** of the ganged read operation (or ganged program verify operation). As shown in FIG. **6**B, in step **2**, both WSGs are closed (or turned off) and SGS**1** and SGS**7** are made open (or turned on). In an embodiment, if the threshold voltage (Vt) of memory cell k (e.g., where k=1 or 2) is higher than a first threshold voltage (e.g., 2V), then the voltage of SN_k (e.g., where k=1 or 2) remains the same. If the threshold voltage of memory cell k is less than the first threshold voltage, then SN_k is discharged via the selected memory cell that is turned 'on'.

[0078] FIG. **6**C illustrates an example step **3** of the ganged read operation (or ganged program verify operation). As shown in FIG. **6**C, in step **3**, the read select gate (RSG) of sense module **1** (i.e., RSG **1**) is made open (or turned on) and RSG **2** of sense module **2** is made closed (or turned off). In an embodiment, if the voltage of SN **1** remains the same, sense module **1** is made open (or is turned on) and the voltage of the global bitline **601** is discharged by sense module **1**. If SN **1** does not remain the same, sense module **1** is closed (or turned off) and the voltage of the global bitline **601** is not discharged by sense module **1** (i.e., the voltage of the global bitline **601** remains the same). As a result, first data stored in the first selected memory cell is transferred from sense module **1** to the page buffer **621** (i.e., the first data of the first selected memory cell is read as part of the ganged read operation or ganged program verify operation).

[0079] FIG. **6**D illustrates an example step **4** of the ganged read operation (or ganged program verify operation). As shown in FIG. **6**D, in step **4**, the read select gate (RSG) of sense module **2** (i.e., RSG **2**) is made open (or turned on) and RSG **1** of sense module **1** is made closed (or turned off). In an embodiment, if the voltage of SN **2** remains the same, sense module **2** is made open (or is turned on) and the voltage of global bitline **601** is discharged by sense module **2**. If SN **2** does not remain the same, sense module **2** is closed (or turned off) and the voltage of the global bitline **601** is not discharged by sense module **2** (i.e., the voltage of the global bitline **601** remains the same). As a result, second data stored in the second selected memory cell is transferred from sense module **2** to the page buffer **621** (i.e., the second data of the second selected memory cell is read as part of the ganged read operation or ganged program verify operation).

[0080] FIG. **7** is a flow diagram of an example method **700** to execute a ganged memory access operation associated with a memory device in a memory sub-system in accordance with some embodiments of the present disclosure. The method **700** is described with reference to FIGS. **1**A-**6**D. The method **700** can be performed by processing logic that can include hardware (e.g., processing device, circuitry, dedicated logic, programmable logic, microcode, hardware of a device, integrated circuit, etc.), software (e.g., instructions run or executed on a processing device), or a combination thereof. In some embodiments, the method **700** is performed by memory access operation manager **134** of FIG. **1**A and FIG. **1**B. Although shown in a particular sequence or order, unless otherwise specified, the order of the processes can be modified. Thus, the illustrated embodiments should be understood only as examples, and the illustrated processes can be performed in a different order, and some processes can be performed in parallel. Additionally, one or more processes can be omitted in various embodiments. Thus, not all processes are required in every embodiment. Other process flows are possible.

[0081] At operation **710**, an operation is initiated. For example, control logic (e.g., memory access operation manager **134**) can initiate the execution of a memory access operation associated with a memory device including a memory block including a first subset of sub-blocks and a second subset of sub-blocks, where a first sense module is coupled to the first subset of sub-blocks and a second sense module is coupled to a second subset of sub-blocks. In an embodiment, the memory access operation can include a ganged program operation (as described in detail above with reference to FIGS. **4** and **5**), a ganged read operation (as described in detail above with reference to FIGS. **6**A-**6**D), or a ganged program verify operation (as described in detail above with reference to FIGS. **6**A-**6**D).

[0082] At operation **720**, first data is transmitted. For example, the control logic can cause transmission, via the first sense module, of first data between a page buffer circuit and a first memory cell of the first subset of sub-blocks. In an embodiment, if the memory access operation initiated in operation **710** is a ganged program operation, the first data is transmitted via the first sense module from the page buffer to the first memory cell (i.e., the first memory cell is programmed with the first data). In an embodiment, if the memory access operation initiated in operation **710** is a ganged read operation or a ganged program verify operation, the first data is transmitted via the first sense module from the first memory cell to the page buffer (i.e., the first data is read out from the first memory cell).

[0083] At operation **730**, second data is transmitted. For example, the control logic can cause transmission, via the second sense module, of second data between the page buffer circuit and a second memory cell of the second subset of sub-blocks. In an embodiment, if the memory access operation initiated in operation **710** is a ganged program operation, the second data is transmitted via the second sense module from the page buffer to the second memory cell (i.e., the second memory cell is programmed with the second data). In an embodiment, if the memory access operation initiated in operation **710** is a ganged read operation or a ganged program verify operation, the second data is transmitted via the second sense module from the second memory cell to the page buffer (i.e., the second data is read out from the second memory cell).

[0084] Advantageously, use of respective sense modules coupled to different sub-blocks of a memory block enables memory cells of the different sub-blocks to be programmed or read concurrently (i.e., as part of the same ganged memory access operation). As a result, the control logic can execute a ganged program operation to program memory cells of different sub-blocks of a memory device concurrently. Furthermore, the control logic can use the multiple sense modules to execute a ganged read operation or ganged program verify operation to read data stored in memory cells of different sub-blocks of a memory device concurrently (i.e., as part of a same ganged read operation or ganged program verify operation). Execution of the ganged memory access operations result in a reduction in the time it takes to perform program operations, read operations, and garbage collection operations, which improves quality of service metrics.

[0085] FIG. **8** illustrates an example machine of a computer system **800** within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, can be executed. In some embodiments, the computer system **800** can correspond to a host system (e.g., the host system **120** of FIG. **1**A) that includes, is coupled to, or utilizes a memory sub-system (e.g., the memory sub-system **110** of FIG. **1**A) or can be used to perform the operations of a controller (e.g., to execute an operating system to perform operations corresponding to memory access operation manager **134** of FIGS. **1**A and **1**B). In alternative embodiments, the machine can be connected (e.g., networked) to other machines in a LAN, an intranet, an extranet, and/or the Internet. The machine can operate in the capacity of a server or a client machine in client-server network environment, as a peer machine in a peer-to-peer (or distributed) network environment, or as a server or a client machine in a cloud computing infrastructure or environment.

[0086] The machine can be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, a switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single machine is illustrated, the term "machine" shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

[0087] The example computer system **800** includes a processing device **802**, a main memory **804** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), etc.), a static memory **806** (e.g., flash memory, static random access memory (SRAM), etc.), and a data storage system **818**, which communicate with each other via a bus **830**.

[0088] Processing device **802** represents one or more general-purpose processing devices such as a microprocessor, a central processing unit, or the like. More particularly, the processing device can be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processing device **802** can also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device **802** is configured to execute instructions **826** for performing the operations and steps discussed herein. The computer system **800** can further include a network interface device **808** to communicate over the network **820**.

[0089] The data storage system **818** can include a machine-readable storage medium **824** (also known as a computer-readable medium, such as a non-transitory computer-readable medium) on which is stored one or more sets of instructions **826** or software embodying any one or more of the methodologies or functions described herein. The instructions **826** can also reside, completely or at least partially, within the main memory **804** and/or within the processing device **802** during execution thereof by the computer system **800**, the main memory **804** and the processing device **802** also constituting machine-readable storage media. The machine-readable storage medium **824**, data storage system **818**, and/or main memory **804** can correspond to the memory sub-system **110** of FIG. **1**A.

[0090] In one embodiment, the instructions **826** include instructions to implement functionality corresponding to memory access operation manager **134** of FIGS. **1**A and **1**B). While the machine-readable storage medium **824** is shown in an example embodiment to be a single medium, the term "machine-readable storage medium" should be taken to include a single medium or multiple media that store the one or more sets of instructions. The term "machine-readable storage medium" shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of the present disclosure. The term "machine-readable storage medium" shall accordingly be taken to include, but not be limited to, solid-state memories, optical media, and magnetic media.

[0091] Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

[0092] It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. The present disclosure can refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage systems.

[0093] The present disclosure also relates to an apparatus for performing the operations herein. This apparatus can be specially constructed for the intended purposes, or it can include a general purpose computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program can be stored in a computer readable storage medium, such as, but not limited to, any type of disk including floppy disks, optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

[0094] The algorithms and displays presented herein are not inherently related to any particular computer or other apparatus. Various general purpose systems can be used with programs in accordance with the teachings herein, or it can prove convenient to construct a more specialized apparatus to perform the method. The structure for a variety of these systems will appear as set forth in the description below. In addition, the present disclosure is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages can be used to implement the teachings of the disclosure as described herein.

[0095] The present disclosure can be provided as a computer program product, or software, that can include a machine-readable medium having stored thereon instructions, which can be used to program a computer system (or other electronic devices) to perform a process according to the present disclosure. A machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). In some embodiments, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium such as a read only memory ("ROM"), random access memory ("RAM"), magnetic disk storage media, optical storage media, flash memory components, etc.

[0096] In the foregoing specification, embodiments of the disclosure have been described with reference to specific example embodiments thereof. It will be evident that various modifications can be made thereto without departing from the broader spirit and scope of embodiments of the disclosure as set forth in the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

What is claimed is:

1. A memory device comprising:

a memory array comprising a memory block, wherein the memory block comprises a first subset of sub-blocks and a second subset of sub-block;

a first sense module coupled to the first subset of sub-blocks;

a second sense module coupled to the second subset of sub-blocks;

a page buffer circuit coupled to the first sense module and the second sense module via a bitline; and

control logic, operatively coupled to the memory block, to perform operations comprising:

initiating a memory access operation associated with the memory block of the memory device;

during the memory access operation, causing transmission, via the first sense module, of first data between the page buffer circuit and a first memory cell of the first subset of sub-blocks; and

during the memory access operation, causing transmission, via the second sense module, of second data between the page buffer circuit and a second memory cell of the second subset of sub-blocks.

2. The memory device of claim **1**, wherein the memory access operation comprises a program operation.

3. The memory device of claim **2**, wherein the first data is transmitted from the page buffer circuit to the first memory cell of the first subset of sub-blocks via the bitline and the first sense module.

4. The memory device of claim **3**, wherein the second data is transmitted from the page buffer circuit to the second memory cell of the second subset of sub-blocks via the bitline and the second sense module.

5. The memory device of claim **1**, wherein the memory access operation comprises one of a read operation or a program verify operation.

6. The memory device of claim **5**, wherein the first data is transmitted from the first memory cell of the first subset of sub-blocks to the page buffer circuit via the first sense module and the bitline.

7. The memory device of claim **6**, wherein the second data is transmitted from the second memory cell of the second subset of sub-blocks to the page buffer circuit via the second sense module and the bitline.

8. A method comprising:

initiating a memory access operation associated with a memory array comprising a memory block of a memory device, wherein the memory block comprises a first subset of sub-blocks and a second subset of sub-blocks;

during the memory access operation, causing transmission, via a first sense module coupled to the first subset of sub-blocks, of first data between a page buffer circuit and a first memory cell of the first subset of sub-blocks; and

during the memory access operation, causing transmission, via a second sense module coupled to the second subset of sub-blocks, of second data between the page buffer circuit and a second memory cell of the second subset of sub-blocks.

9. The method of claim **8**, wherein the memory access operation comprises a program operation.

10. The method of claim **9**, wherein the first data is transmitted from the page buffer circuit to the first memory cell of the first subset of sub-blocks via a bitline and the first sense module.

11. The method of claim **10**, wherein the second data is transmitted from the page buffer circuit to the second memory cell of the second subset of sub-blocks via the bitline and the second sense module.

12. The method of claim **8**, wherein the memory access operation comprises one of a read operation or a program verify operation.

13. The method of claim **12**, wherein the first data is transmitted from the first memory cell of the first subset of sub-blocks to the page buffer circuit via the first sense module and a bitline.

14. The method of claim **13**, wherein the second data is transmitted from the second memory cell of the second subset of sub-blocks to the page buffer circuit via the second sense module and the bitline.

15. A non-transitory computer-readable medium comprising instructions, which when executed by a processing device, cause the processing device to perform operations comprising:

initiating a memory access operation associated with a memory block of a memory device, wherein the memory block comprises a first subset of sub-blocks and a second subset of sub-blocks;

during the memory access operation, causing transmission, via a first sense module coupled to the first subset of sub-blocks, of first data between a page buffer circuit and a first memory cell of the first subset of sub-blocks; and

during the memory access operation, causing transmission, via a second sense module coupled to the second subset of sub-blocks, of second data between the page buffer circuit and a second memory cell of the second subset of sub-blocks.

16. The non-transitory computer-readable medium of claim **15**, wherein the memory access operation comprises a program operation.

17. The non-transitory computer-readable medium of claim **16**, wherein the first data is transmitted from the page buffer circuit to the first memory cell of the first subset of sub-blocks via a bitline and the first sense module; and wherein the second data is transmitted from the page buffer circuit to the second memory cell of the second subset of sub-blocks via the bitline and the second sense module.

18. The non-transitory computer-readable medium of claim **15**, wherein the memory access operation comprises one of a read operation or a program verify operation.

19. The non-transitory computer-readable medium of claim **18**, wherein the first data is transmitted from the first memory cell of the first subset of sub-blocks to the page buffer circuit via the first sense module and a bitline.

20. The non-transitory computer-readable medium of claim **19**, wherein the second data is transmitted from the second memory cell of the second subset of sub-blocks to the page buffer circuit via the second sense module and the bitline.

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